

Project #12: Repair JBLM Railhead

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Reference Document: ERDC/GSL APB ARR-20-06, "Railroad Track Inspection and Repair Recommendations, Joint Base Lewis-McChord, Washington" (Dated: March 2021)

Scope: The attached "Railroad Track Inspection and Repair Recommendations" report is provided to all prospective Contractors for review and to serve as the foundational basis scope understanding.

The Contractor shall be responsible for conducting a thorough review of the Report and addressing all specified defects and repair requirements. The recommendations contained within the Report are to be considered the minimum requirements to be met for project completion.

Joint Base Lewis-McChord (JBLM) encourages Contractors to propose and incorporate innovative solutions that may enhance the quality, efficiency, or longevity of the railroad infrastructure. This includes, but is not limited to, the utilization of novel materials, the implementation of advanced or streamlined processes, and the proposal of optimized project timelines.



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Railroad Track Inspection and Repair Recommendations, Joint Base Lewis- McChord, Washington

Andy Anderton, Jeremy Beasley

March 2021



Geotechnical and Structures Laboratory

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Preface

This report presents the results of an assessment of the condition of railroad track at Joint Base Lewis-McChord, WA. Users of information from this report include the personnel at Joint Base Lewis-McChord Directorate of Public Works (DPW), train master (locomotive supervisor), and yard master (switchmen and brakemen supervisor).

The Office of the Deputy Chief of Staff (DCS), G-9 is the Headquarters, Department of the Army proponent for the Army Railroad Evaluation Program. The Army Railroad Evaluation Program is sponsored by the Army Dams and Transportation Infrastructure Program (ADTIP) of the Headquarters, Installation Management Command (IMCOM), Arlington, VA. The IMCOM provided funding for this investigation. Questions should be directed to Michael Andres, ADTIP Program Manager, Email: michael.r.andres.civ@mail.mil, mailing address is as follows:

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Personnel of the U.S. Army Engineer Research and Development Center (ERDC), Geotechnical and Structures Laboratory (GSL), Vicksburg, MS, performed the assessment. The findings and recommendations presented in this report are based on the detailed visual and ultrasonic internal flaw inspections at Joint Base Lewis-McChord, conducted during November 2020 by Andy Anderton, Jeremy Beasley, and Mike Crawford of the APB, GSL. Andy Anderton prepared this report under the supervision of Mrs. Anna Jordan, Chief, APB; Mr. Justin S. Strickler, Chief, Engineering Networks and Materials Division; Mr. Charles W. Ertle, Deputy Director, GSL; and Mr. Bartley P. Durst, Director, GSL. At the time of publication of this report, COL Teresa A. Schlosser was the Commander and Executive Director of ERDC, and Dr. David W. Pittman was the Director.

Unit Conversion Factors

Multiply	By	To Obtain
degrees Fahrenheit	$(F-32)/1.8$	degrees Celsius
feet	0.3048	meters
inches	0.0254	meters
miles (U.S. statute)	1,609.347	meters
miles per hour	0.44704	meters per second
pounds (mass) per yard	0.496054648	kilograms per meter
tons (2,000 pounds, mass)	907.1847	kilograms

Executive Summary

Personnel of the U.S. Army Engineer Research and Development Center (ERDC), Vicksburg, MS, performed on-site assessments of railroad facilities at Joint Base Lewis-McChord, WA, during November 2020. A schematic of Joint Base Lewis-McChord tracks can be found in Figure ES-1. The condition of the railroad facilities was determined using visual and ultrasonic internal flaw inspections to identify and delineate apparent defects.

The results of the railroad inspection reveal the following:

- a. Table ES-1 lists the close-to-traffic defects found during this inspection.
- b. Of the 35 track segments inspected, the operating restrictions of 30 segments are green (no traffic restrictions), 0 are amber (restricted operations), and 5 are red (close-to-traffic). These operating restrictions are shown in Figures ES-1 through ES-7 and listed in Table 3-2.
- c. The individual track structure condition indices (TSCI) for all track segments ranged from 98 - 100. The TSCI, listed in Table 3-4, is computed by RAILER. It can be used to assess the overall condition of the track segment and as a comparison with future inspections.
- d. The estimated cost of maintenance and repair to render all track segments defect-free at Joint Base Lewis-McChord is approximately [REDACTED]. Repair costs given in this report are estimated national averages and do not allow for variation among different geographical locations.
- e. Appendix A has photos of example defects found during the inspection.
- f. Appendix B is an inventory of railroad facilities at Joint Base Lewis-McChord.
- g. Defects, recommended rehabilitation procedures, and cost estimates associated with the railroad track facilities at Joint Base Lewis-McChord are listed in Appendix C.
- h. Appendix D contains any available information on railroad system work and property activity documentation for Joint Base Lewis-McChord.
- i. Appendix E contains a summary of the Joint Base Lewis-McChord Railroad Deficiency Report as stated by HQ IMCOM.

Critical findings

Critical findings for the detailed visual and ultrasonic internal flaw inspections are defects that place close-to-traffic restrictions on track segments as specified by the UFC 4-860-03 (Headquarters, Departments of the Army, the Navy, and the Air Force 2008). These defects should be corrected immediately before operations over the tracks are allowed. Table ES-1 lists the close-to-traffic defects found at Joint Base Lewis-McChord during the November 2020 inspection.

Table ES-1. List of close-to-traffic defects found at Joint Base Lewis-McChord, November 2020.

Defect #	Track Segment	Station (ft)	Component	ID	Defect Description	Meas (in.)	QTY	UM	Work Action	Comments
1	Lc 10	6+45 - 7+61	Turnouts	4	Switch Rods, Loose		1	EA	Tighten/Adjust Switch Rod	
2	Lc 10	16+86 - 18+02	Turnouts	11	Switch Rods, Loose		1	EA	Tighten/Adjust Switch Rod	
3	Lc 10	43+57 - 44+73	Turnouts	27	Switch Rods, Loose		1	EA	Tighten/Adjust Switch Rod	
4	Madigan Spur	0+00 - 47+21	Ties		Isolated Defective Tie Cluster (5 Ties)		1	EA	Install/Replace Five Ties (7"x9"x8.5')	
5	Main	3+20 - 4+36	Turnouts	3	Switch Rods, Loose		1	EA	Tighten/Adjust Switch Rod	
6	Main	36+01 - 37+17	Turnouts	21	Switch Rods, Loose		1	EA	Tighten/Adjust Switch Rod	
7	Track 6	8+27 - 9+43	Turnouts	30	Switch Rods, Loose		1	EA	Tighten/Adjust Switch Rod	
8	Lead 2	30+55 - 31+53	Turnouts	45	Point Rails (Right), Improper Type		1	EA	Install or Replace Point Rail	
9	Lead 2	30+55 - 31+53	Turnouts	45	Switch Rods, Loose		1	EA	Tighten/Adjust Switch Rod	

Inspection meetings

A pre-inspection meeting was held on November 13, 2020 at Joint Base Lewis-McChord, WA. The purpose of this meeting was to discuss the purpose and processes of the inspection with the installation managers. Items discussed during the pre-inspection briefing included:

- a. Communication between the inspection team and train master.
- b. Layout and primary uses of railroad track facilities.
- c. Locations of known distresses on railroad track facilities.
- d. Planned rail movements during the course of the inspection.

Table ES-2 summarizes the attendees present at the pre-inspection meeting, including their contact information.

Table ES-2. Pre-inspection briefing attendance list.

Name	Organization	Phone	Email
[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]	[REDACTED]	[REDACTED]

A post-inspection briefing was held on November 23, 2020 to provide Joint Base Lewis-McChord, WA, personnel with a summary of the findings from the railroad inspection. Details presented at the post-inspection briefing included the location of critical defects and the location of rail closures and/or restrictions. The post-inspection meeting also included time for questions from personnel about the inspection results and recommended follow-on actions.

Table ES-3 lists the attendees present at the post-inspection meeting, including their contact information.

Table ES-3. Post-inspection briefing attendance list.

[REDACTED]

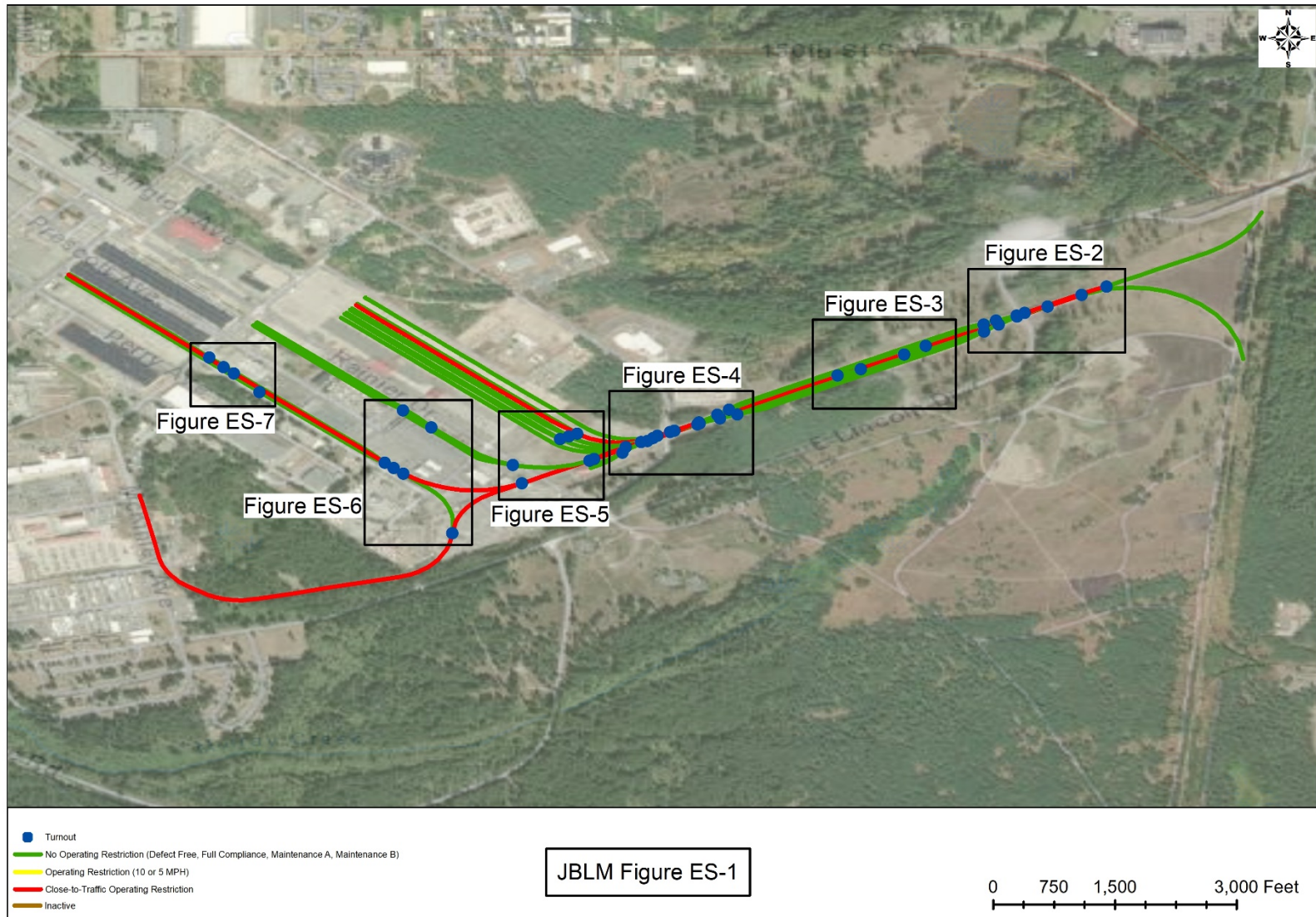


Figure ES-1. Overview of Joint Base Lewis-McChord track network.

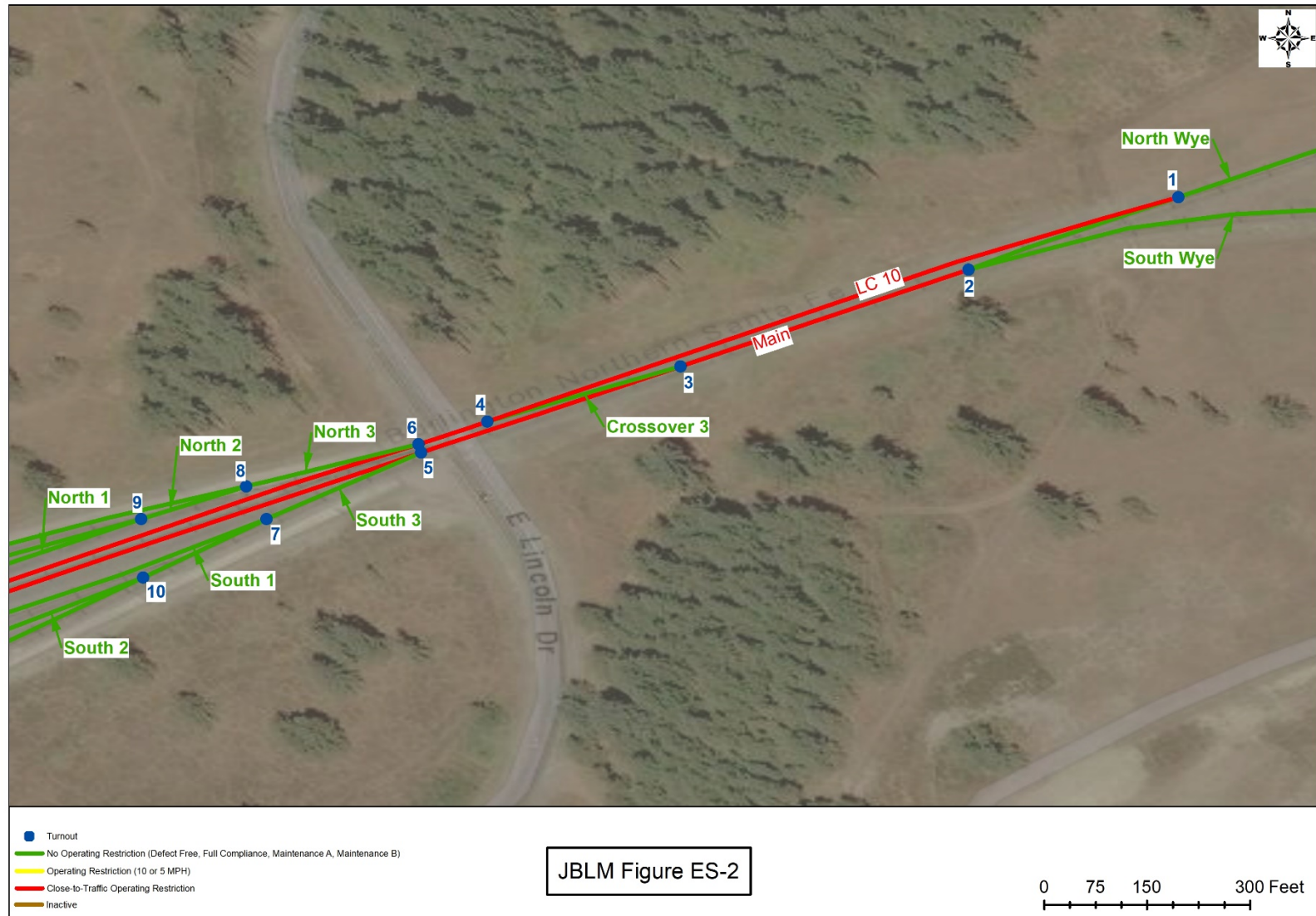


Figure ES-2. Operating restrictions for North 1, North 2, North 3, South 1, South 2, South 3, Crossover 3, Main, LC 10, North Wye and South Wye.

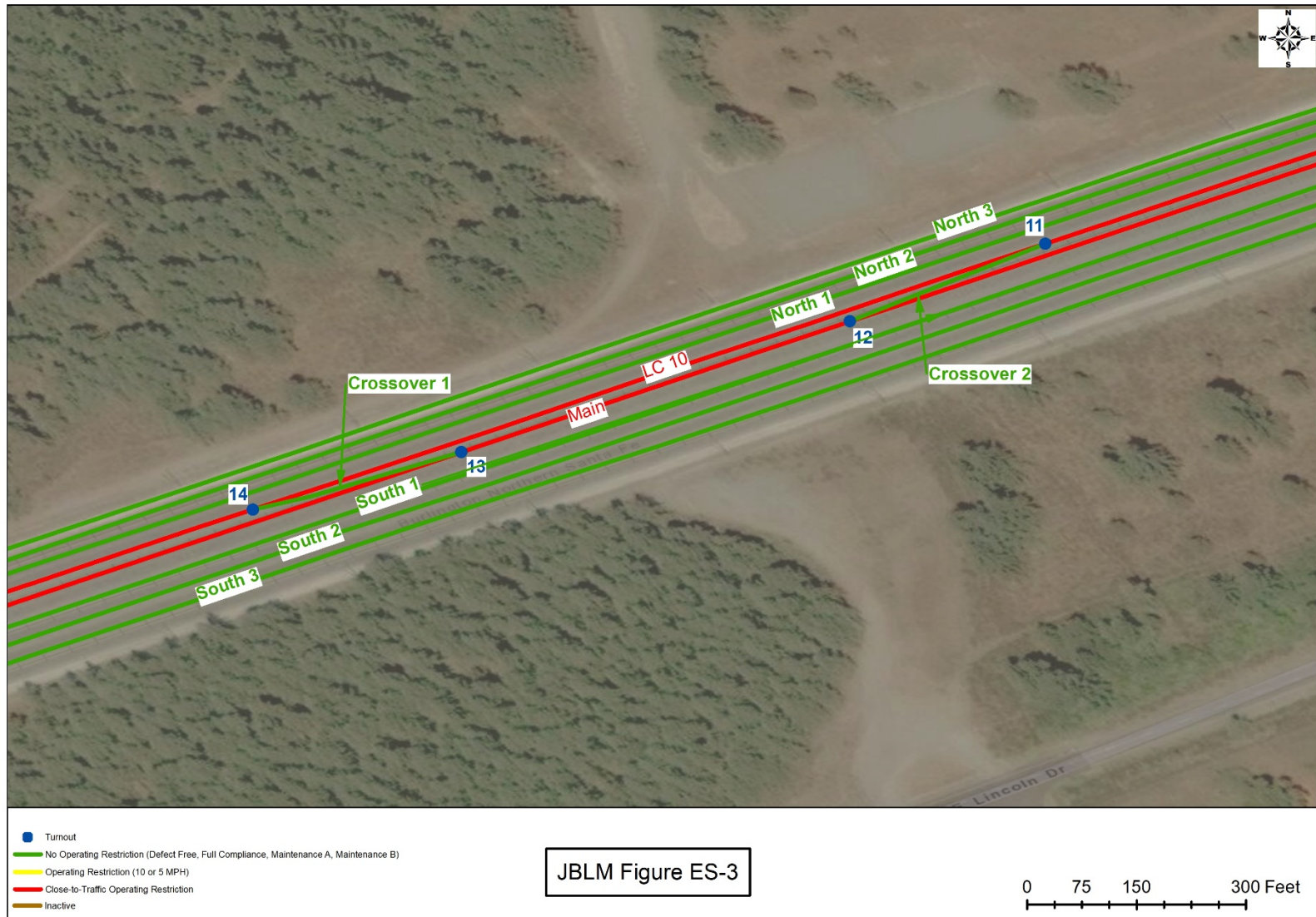


Figure ES-3. Operating restrictions for North 1, North 2, North 3, South 1, South 2, South 3, Crossover 1, Crossover 2, Main and LC 10.

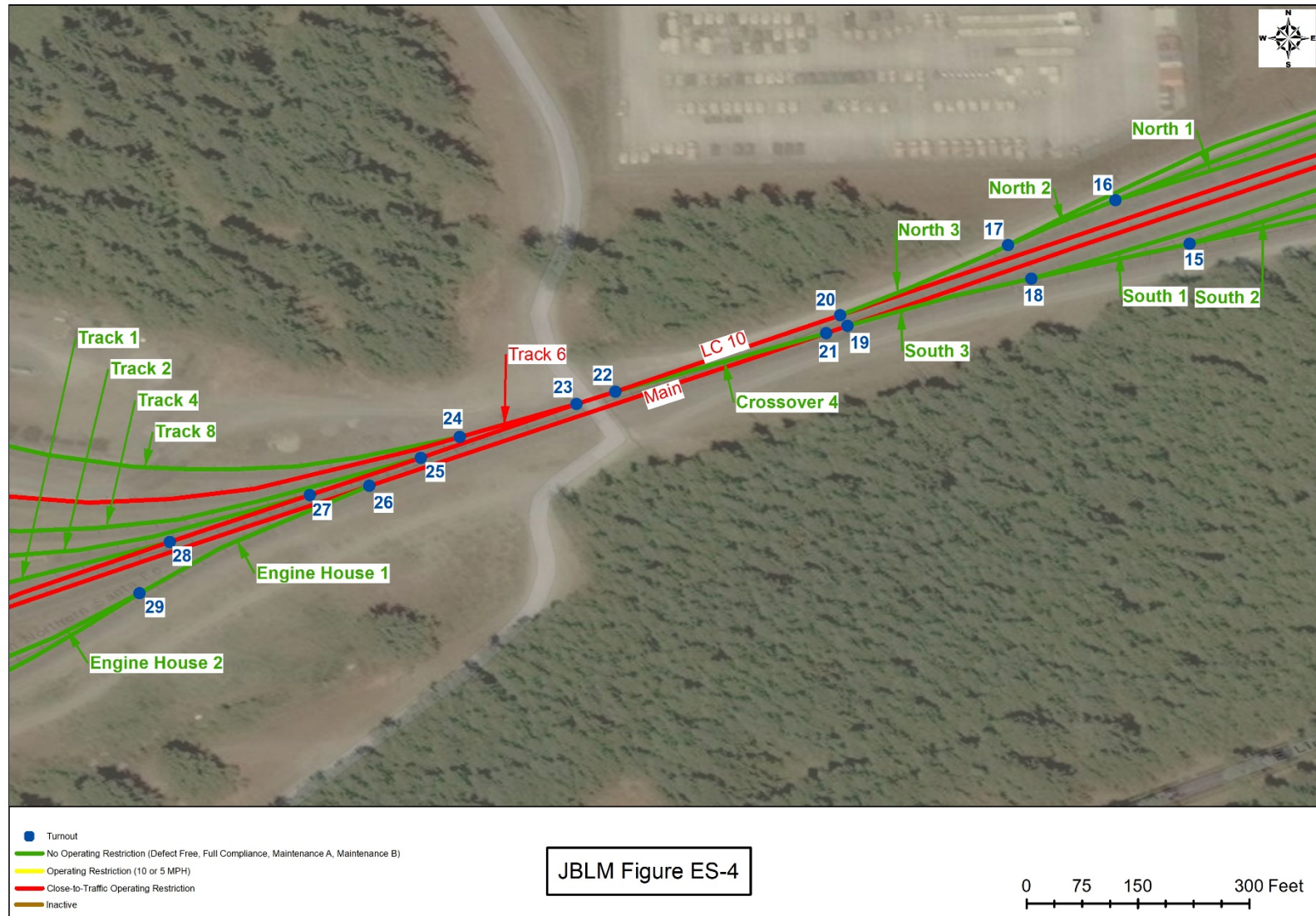


Figure ES-4. Operating restrictions for North 1, North 2, North 3, South 1, South 2, South 3, Main, LC 10, Track 1, Track 2, Track 4, Track 6, Track 8, Engine House 1 and Engine House 2.

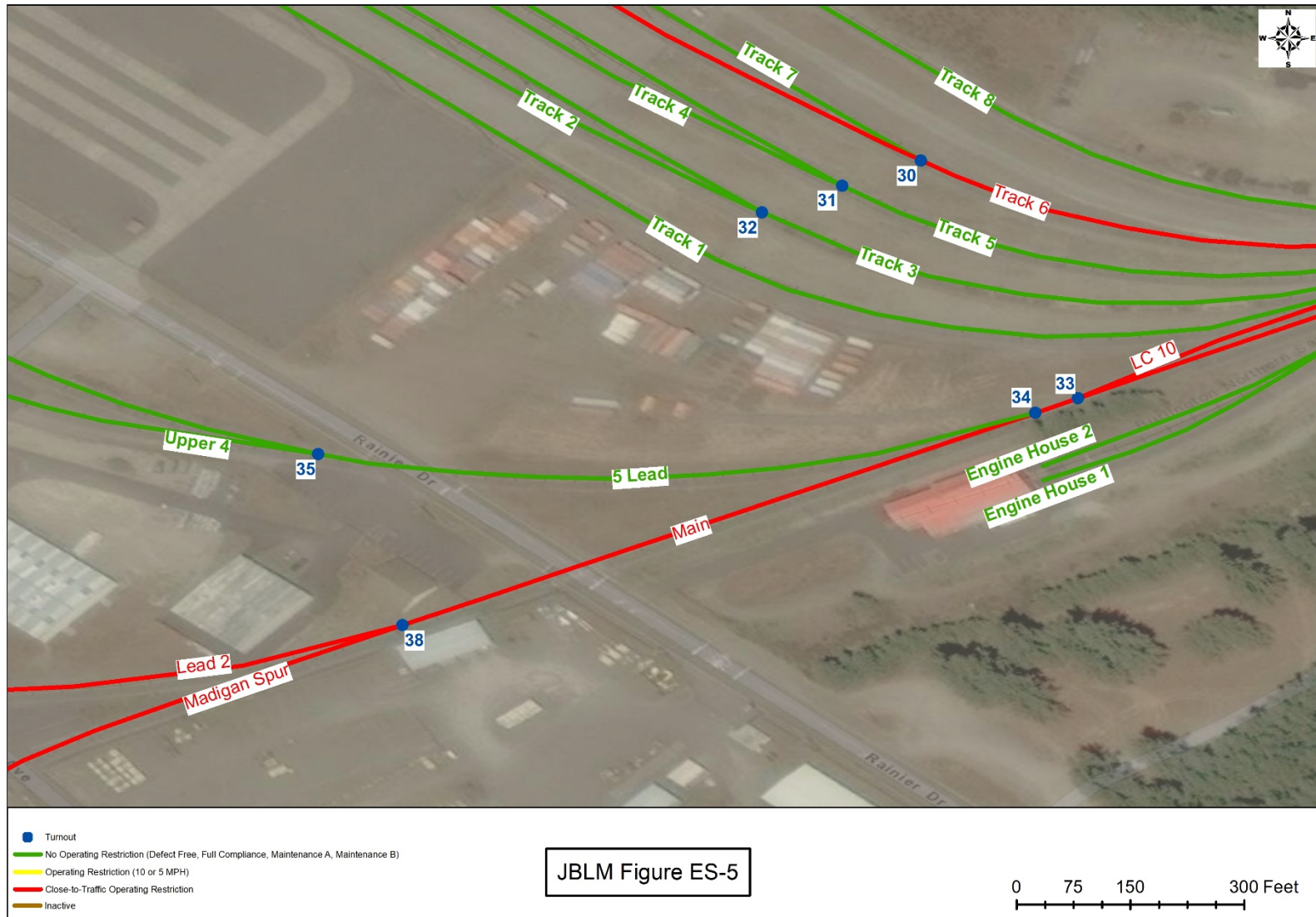


Figure ES-5. Operating restrictions for Main, LC 10, Track 1, Track 2, Track 3, Track 4, Track 5, Track 6, Track 7, Track 8, Engine House 1, Engine House 2, 5 Lead, Upper 4, Lead 2 and Madigan Spur.

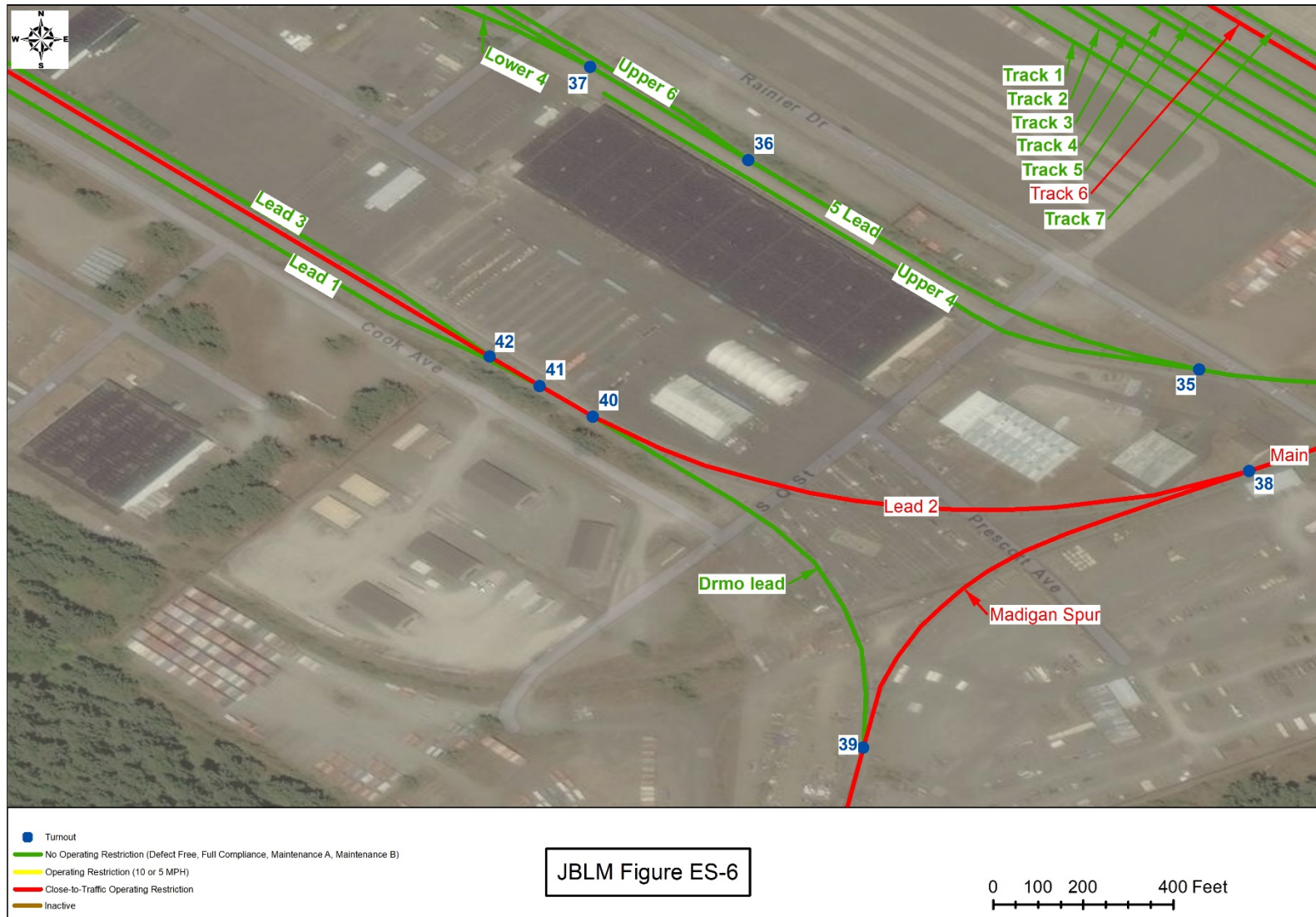


Figure ES-6. Operating restrictions for Main, Track 1, Track 2, Track 3, Track 4, Track 5, Track 6, Track 7, 5 Lead, Upper 4, Upper 6 Lower 4, Lead 2, Lead 1, Lead 3, Drmo lead and Madigan Spur.

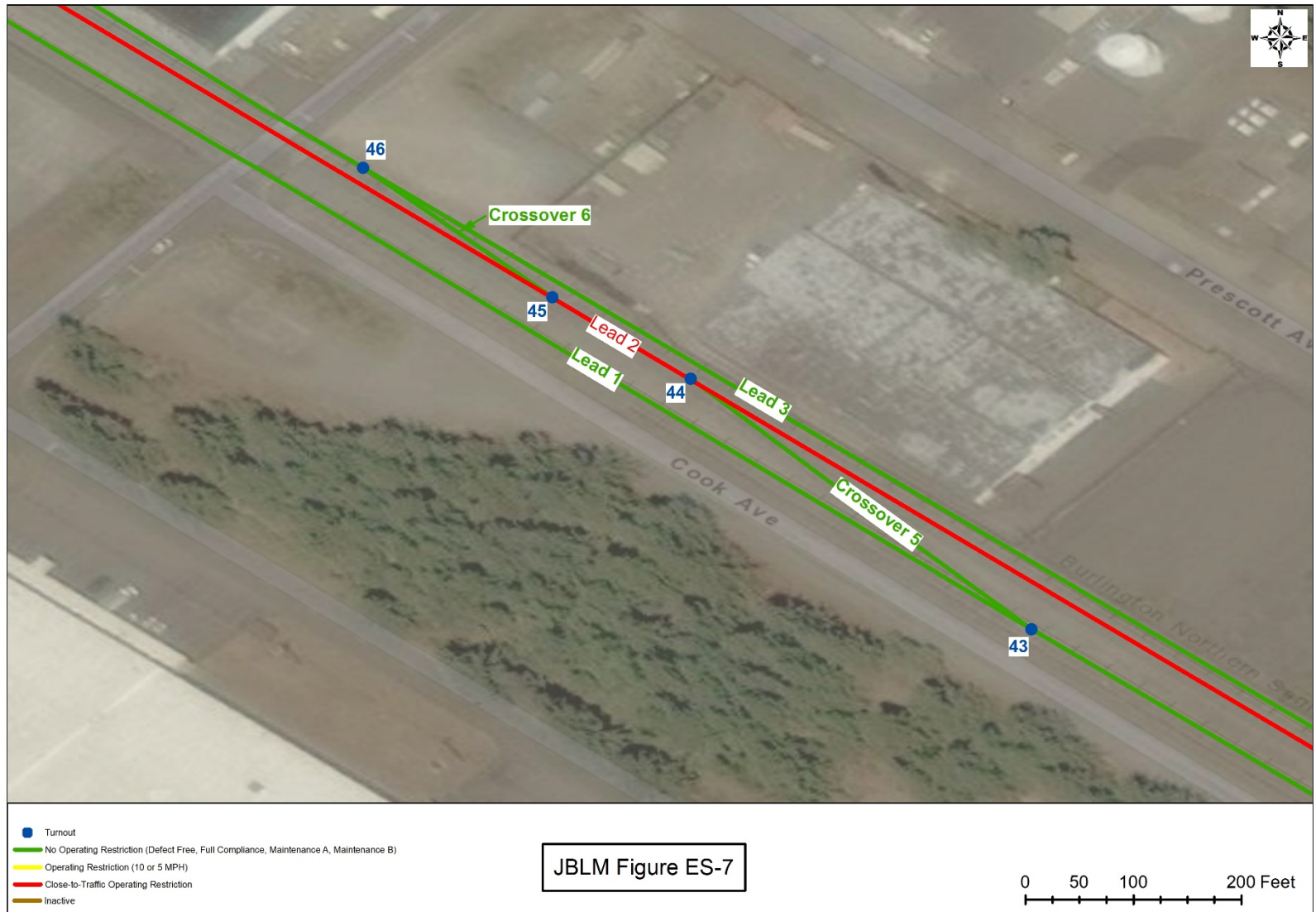


Figure ES-7. Operating restrictions for Lead 2, Lead 1, Lead 3, Crossover 5 and Crossover 6.

1 Introduction

Background

The Installation Management Command (IMCOM) tasked the U.S. Army Engineer Research and Development Center (ERDC) to evaluate the railroad track facilities at Joint Base Lewis-McChord during November 2020. Railroad facilities at Joint Base Lewis-McChord were last inspected by the ERDC in 2015.

Objective and scope

The objectives of this investigation were to determine the condition of the railroad track facilities and to identify maintenance and repair requirements for each railroad track segment. Specific objectives were to:

- a. Perform detailed visual and ultrasonic internal flaw inspections of all railroad tracks at Joint Base Lewis-McChord in accordance with UFC 4-860-03 (Headquarters, Departments of the Army, the Navy, and the Air Force 2008).
- b. Mark defective ties, rails, joint bars, and other track material (OTM) that need replacement.
- c. Implement the RAILER software for analysis of the rail conditions, develop maintenance and repair recommendations, and provide estimates of quantities and costs associated with the maintenance and repair required to meet defect-free maintenance standards as defined in UFC 4-860-03 (Headquarters, Departments of the Army, the Navy, and the Air Force 2008).
- d. Identify the condition of track segments according to the operating restrictions and track structure condition indices (TSCI) computed by RAILER.
- e. Make recommendations regarding the surface and alignment of the track, grade crossing enhancements, turnout geometry, track layout, and other items to improve the overall operating conditions of the railroad facilities.
- f. Identify hazardous conditions that require closure or restriction of tracks.

The results of this study can be used to:

- a. Correct critical defects that require closure or restriction of tracks to traffic to meet defect-free maintenance standards as defined in UFC 4-860-03 (Headquarters, Departments of the Army, the Navy, and the Air Force 2008).
- b. Assist in identifying and forecasting maintenance and repair work.
- c. Determine the repairs needed to sustain current levels of railroad track operations.

Chapter 2 of this report contains a summary of the results of the detailed visual and ultrasonic internal flaw inspections, including critical defects requiring immediate repair. Chapter 3 includes operating restrictions and the overall condition index for each track segment. Chapter 4 summarizes the estimated costs to bring all of the tracks back to full compliance. Conclusions and recommendations are summarized in Chapter 5. Appendix A has photographs of examples of defects found during the inspection. Appendix B lists the current inventory of the Joint Base Lewis-McChord railroad facilities as determined by this inspection. Appendix C lists all defects found during the inspection by track segment, the rehabilitation procedures recommended for the railroad track facilities at Joint Base Lewis-McChord, and their estimated costs.

Description of track network

Joint Base Lewis-McChord has 15.24 measured miles of active track. There is also an estimated 2 miles of inactive track that has been disconnected from active track. There is also additional track covered by vegetation and debris that is unable to be measured. The estimated mileage was gathered using measurement tools on satellite imagery. Joint Base Lewis-McChord real property records show a total of 25.2 miles of active track and 7 miles of inactive track. The track network consists of two wye tracks that connect to the serving railroad (BNSF), two classification yards, storage tracks, engine house tracks and loading ramp tracks. The two classification yards are both contained behind gates, with one yard's tracks ending at concrete end ramps.

2 Results and Recommendations

Overview of current conditions

This chapter discusses the results of the evaluation of the current railroad network condition at Joint Base Lewis-McChord. Defects found for each component are discussed. All defects that have been found and noted in the report should be repaired or replaced in accordance with UFC 4-860-03 (Headquarters, Departments of the Army, the Navy, and the Air Force 2008).

During the inspection, the ERDC inspection team implemented a custom designed, GPS mobile application to digitally record inventory and inspection items of the track system. The Electronic Railroad Inspection Database System (ERIDS) stores data on a GPS device which synchronizes with a local internet server. Various post processing tools enable the user to increase the accuracy and stationing of the track structure, track components, and defects. This data is housed in an Enterprise GeoDatabase that can be exported and consumed by RAILER software for detailed report generation.

Inventory and distress data for the inspection are managed using the RAILER software developed by the ERDC, Construction Engineering Research Laboratory (CERL), Champaign, IL (U.S. Army ERDC 2008). RAILER is utilized to develop maintenance and repair recommendations and to provide estimates of quantities and costs associated with the maintenance and repair required to meet defect-free standards as defined in UFC 4-860-03 (Headquarters, Departments of the Army, the Navy, and the Air Force 2008).

Defects are categorized into maintenance and safety/operating standards. Maintenance standards provide limits for defects and should be used as a guideline to determine when maintenance is recommended. Although maintenance standards do not impose any traffic restrictions on the track, repair is still recommended so that the defect does not deteriorate and cause a safety/operating restriction. Maintenance standards are subdivided into A and B categories. Maintenance A standards are recommended for Category A track, and Maintenance B standards are recommended for both Category A and B track.

Safety/operating standards provide the minimum safe operating limits for defects. Safety/operating standards are subdivided into 10 MPH, 5 MPH, and close-to-traffic operating restrictions.

Photos of example distresses at Joint Base Lewis-McChord can be found in Appendix A. Inventory is listed in Appendix B. All defects found during the inspection are listed in Appendix C alphabetically by track segment name.

Actions Since Last Inspection

Crosstie replacement and other routine maintenance was performed and the track network was brought in to full compliance.

Rail

All rail inspected at Joint Base Lewis-McChord is 115 lb/yd rail and is full compliance. Prior to the detailed track inspection, an internal flaw analysis of the track was conducted. Internal flaw analysis was accomplished with a hi-rail truck equipped with an ultrasonic testing unit (NORDCO Testing and Rail Services) and operated by an ERDC personnel certified in Level 2 UT inspection. This ultrasonic testing device was able to test both rails at a rate of approximately 5 MPH. The testing apparatus is capable of detecting the following defects: (1) rail head defects and rail web defects inside and outside the joint bar area; (2) centrally located transverse defects in the rail head and elsewhere in the rail; (3) bolt hole cracks; (4) head and web separations; (5) vertical/ longitudinal defects in the head or web; (6) torch cut bolt holes; and (7) torch cut rail ends. No internal rail defects were found during this inspection.

Fastenings and other track materials (F&OTM)

The fasteners and materials inspected within the track network are in full compliance, with the exception of some loose or missing bolts and small rail end mismatches. These defects are most likely a result of the longitudinal movement during thermal expansion and contraction of the rail. This movement can be minimized with the use of rail anchors. Rail anchors are discussed in the UFC 4-860-03, Section 6-5, and it is recommended to install a minimum of eight anchors, four in each direction (boxed), per 39-foot rail or 16 anchors per 80-foot rail.

Ties

There are approximately 45,720 crossties in the Joint Base Lewis-McChord track network with only 230 crossties, or 0.5%, were found to be defective. This is well below the normal average of 1-3 percent of defective crossties found in a nominal condition track network. This can be credited to an excellent tie replacement program. One isolated 5 crosstie defective cluster was found at the end of Madigan Spur, which is a closed- to-traffic defect.

Ballast

The ballast of any track network plays a significant role in proper drainage which further increases the usable life of crossties. Ballast is a select angular material placed on the subgrade to provide adequate drainage, restrain the track laterally, longitudinally, and vertically under the dynamic loads imposed by trains and the thermal stresses induced in the rails by changing temperature, and distribute the load of the track and trains to prevent overstressing the subgrade. There are sections of ballast fouled with dirt and other fine materials throughout the track network. Pumping crossties were found on Track 1, Track 2 and Track 4. Pumping is the periodical up and down movement of crossties under a train movement. This can cause the deterioration of crossties at a much faster rate. Pumping occurs more often in wet climates and is considered a Maintenance A defect.

Drainage/subgrade

Drainage is in good condition, both from the track and through culverts. All culverts and drain pipes were at least 75% clear at the time of the inspection.

Note: While the overall drainage conditions were observed and general recommendations made, detailed determination of drainage repairs such as sizing drainage structures, running levels, etc., and road-crossing grade changes were beyond the scope of this inspection.

Vegetation

Vegetation is growing in the drainage ditches along the sides of the track but has been controlled on all tracks and is in full compliance.

All vegetation should be sprayed 8 ft to both sides of the track centerline. Spraying a larger area will be more effective at controlling vegetation. Areas that have been encroached by vegetation should be sprayed with herbicide in accordance with AR 420-1, TM 5-629, and TM 5-630.

Turnouts

The turnouts of any track network are considered the most critical portion. The majority of turnouts were found to be in full compliance with the exception of some turnouts on tracks LC 10, Main, Track 6 and Lead 2. These turnouts were found to have loose switch rods, which is a closed-to-traffic defect. Turnout 45, on Lead 2, will need a new point rail to replace the current point rail or the point rail should be ground to compliance. This will help keep wheels from climbing the point and riding out of the rails resulting in a derailment.

Road crossings

All road crossings at Joint Base Lewis-McChord are in full compliance. Road crossings were recently upgraded and installed and remain in good condition. It is important to ensure that all safety signs, lights, and road markings are clearly visible. All crossing signs, signals, and/or warnings shall conform to the requirements of the Manual on Uniform Traffic Control Devices (MUTCD), Part VIII, "Traffic Control Systems for Railroad-Highway Grade Crossings".

Track geometry

Track geometry at Joint Base Lewis-McChord is in full compliance. It is recommended that all tracks be resurfaced and aligned every 5 years as the condition of the track system may change on jointed track due to thermal expansion and contraction of the tracks.

Resurface and align any tracks that are not within tolerances stated in UFC 4-860-03 (Headquarters, Departments of the Army, the Navy, and the Air Force 2008). Although not required, a minimum of 1-in. superelevation is recommended on any curve greater than 4 degrees for best results.

Bridges

For track inspection purposes, bridges are only inspected from the stringers up which includes the bridge ties and the components resting on

them. There are no bridges in the Joint Base Lewis-McChord track network.

3 Condition Levels

Track segment conditions are quantified using two different methods, the operating restrictions and Track Structure Condition Indices (TSCI) computed by RAILER.

Operating restrictions

The operating restrictions are based on the safety and maintenance standards defined in UFC 4-860-03 (Headquarters, Departments of the Army, the Navy, and the Air Force 2008). Table 3-1 defines the operating restrictions.

The operating restrictions for all of the Joint Base Lewis-McChord tracks are illustrated in Figures ES-2 through ES-7 and are also listed in Table 3-2. Close-to-traffic defects are listed in Table ES-1 in the Executive Summary.

Table 3-1. Definition of operating restrictions.

Operating Restriction Color	Track Standard(s)	Definition ¹
Brown	Inactive	Track is not in use.
GREEN	Defect Free Full Compliance Maintenance A Maintenance B	Track that meets all the requirements of the standards. Track at this level has, at most, only minor defects and should be fully capable of handling all train operations within the operating rules of the installation. Routine maintenance is required to maintain this condition level, and repair work will generally be only minor.
AMBER	10 MPH 5 MPH	Track has defects serious enough to make it unsuitable for operations greater than 10 MPH, essentially "normal yard speed." Based on the inspector's judgment, track which has defects serious enough to make it unsuitable for operations at 10 MPH may be reduced to 5 MPH. Essentially, a train may be allowed to "crawl" over the track. This level is intended as a warning zone to indicate that the track is approaching a condition that will require removal from service.
RED	Close-to-Traffic	Track has defects serious enough to require removal from service. At this level, the operation of trains over the track is generally considered hazardous. Operations shall not be allowed until repairs are completed. However, such track may be used under written permit from the track management authority and in the presence of a track inspector. During operations over the closed portion of track, the track inspector shall have visual contact with the train operator at all times and be prepared to signal the train crew to stop.
¹ Definitions are taken from UFC 4-860-03 (Headquarters, Departments of the Army, the Navy, and the Air Force 2008)		

Table 3-2. Operating restrictions for Joint Base Lewis-McChord segments.

Track Segment	Segment Stations (ft)	Standard	Operating Restriction Color
5 Lead	0+00 - 34+50	Maintenance B	GREEN
Crossover 1	0+00 - 2+11	Defect Free	GREEN
Crossover 2	0+00 - 2+12	Defect Free	GREEN
Crossover 3	0+00 - 2+08	Maintenance A	GREEN
Crossover 4	0+00 - 2+09	Defect Free	GREEN
Crossover 5	0+00 - 3+17	Maintenance A	GREEN
Crossover 6	0+00 - 1+69	Maintenance A	GREEN
Drmo Lead	0+00 - 8+92	Maintenance A	GREEN
Engine House 1	0+00 - 5+86	Maintenance A	GREEN
Engine House 2	0+00 - 3+18	Defect Free	GREEN
Lc 10	0+00 - 48+04	Close to Traffic	RED
Lead 1	0+00 - 36+15	Maintenance B	GREEN
Lead 2	0+00 - 47+88	Close to Traffic	RED
Lead 3	0+00 - 35+27	Maintenance A	GREEN
Lower 4	0+00 - 16+40	Maintenance B	GREEN
Madigan Spur	0+00 - 47+21	Close to Traffic	RED
Main	0+00 - 52+47	Close to Traffic	RED
North 1	0+00 - 23+84	Maintenance B	GREEN
North 2	0+00 - 26+16	Maintenance B	GREEN
North 3	0+00 - 29+82	Maintenance B	GREEN
North Wye	0+00 - 18+63	Maintenance B	GREEN
South 1	0+00 - 26+13	Maintenance B	GREEN
South 2	0+00 - 23+13	Maintenance B	GREEN
South 3	0+00 - 29+83	Maintenance B	GREEN
South Wye	0+00 - 18+40	Maintenance A	GREEN
Track 1	0+00 - 30+06	Maintenance A	GREEN
Track 2	0+00 - 30+99	Maintenance B	GREEN
Track 3	0+00 - 23+79	Maintenance A	GREEN
Track 4	0+00 - 31+63	Maintenance B	GREEN
Track 5	0+00 - 24+11	Maintenance B	GREEN
Track 6	0+00 - 32+71	Close to Traffic	RED
Track 7	0+00 - 24+45	Maintenance A	GREEN
Track 8	0+00 - 30+96	Maintenance A	GREEN
Upper 4	0+00 - 11+14	Maintenance A	GREEN
Upper 6	0+00 - 19+56	Maintenance A	GREEN

Condition index

The condition index was developed by the ERDC, Construction Engineering Research Laboratory (CERL), Champaign, IL. The condition index is a separate indicator of track condition from the operating restriction. The condition index ranges from 0 to 100 and is based on the inspection results. The condition index serves two major purposes:

- a. It describes the overall condition and reflects the maintenance and rehabilitation required.
- b. It documents the condition of the track segment over time to determine the deterioration rate.

While the operating restriction deals strictly with the track standards and illustrates the compliance of each track segment to the safety/operation and maintenance standards described in the UFC 4-860-03, the condition index is a broader measure of the track condition. One defect could result in a close-to-traffic operating restriction for a track segment, even though the track is otherwise in good condition. On the other hand, if there are many defects throughout the track segment, the resulting condition index would be poor even though none of these defects fall below the full compliance level of the UFC 4-860-03 standards. The condition index can serve as a measure of track quality from inspection to inspection as well as a measure of maintenance requirements.

The overall condition index, TSCI, is the average of the condition indices of three track component groups: rail, joints, and fastenings (RJCI); crossties and switch ties (TCI); and ballast, subgrade, and roadway (BSCI). Each track-component group index reflects its capability of supporting typical military, short line, or industrial traffic. These indices are determined from the type, severity, and density of the distresses found during the inspection.

The TSCI can be placed into one of seven condition categories. Table 3-3 lists these categories and the TSCI ranges to which they correspond. This table also gives the general description for each category.

RAILER was used to calculate the condition indices. Table 3-4 lists the condition indices for all track segments at Joint Base Lewis-McChord.

More information about development of the condition indices is given in Uzarski (1993a, 1993b).

Table 3-3. Track structure condition index (TSCI) guidelines.

TSCI	Category	Condition Description ¹
86-100	Excellent	Very few defects. Track function is not impaired. No immediate work action is required, but routine or preventative maintenance could be scheduled for accomplishment.
71-85	Very Good	Minor deterioration. Track function is not impaired. No immediate work action is required, but routine or preventive maintenance could be scheduled for accomplishment.
56-70	Good	Moderate deterioration. Track function may be somewhat impaired. Routine maintenance or minor repair may be required.
41-55	Fair	Significant deterioration. Track function is impaired, but not seriously. Routine maintenance or minor repair is required.
26-40	Poor	Severe deterioration over a small percentage of the track. Less severe deterioration may be present in other portions of the track. Track function is seriously impaired. Major repair is required.
11-25	Very Poor	Critical deterioration has occurred over a large percentage or portion of the track. Less severe deterioration may be present in other portions of the track. Track is barely functional. Major repair or less than total reconstruction is required.
0-10	Failed	Extreme deterioration has occurred throughout nearly all or the entire track. Track is no longer functional. Major repair, complete restoration, or total reconstruction is required.
¹ Taken from FM-93-13 (Uzarski 1993a).		

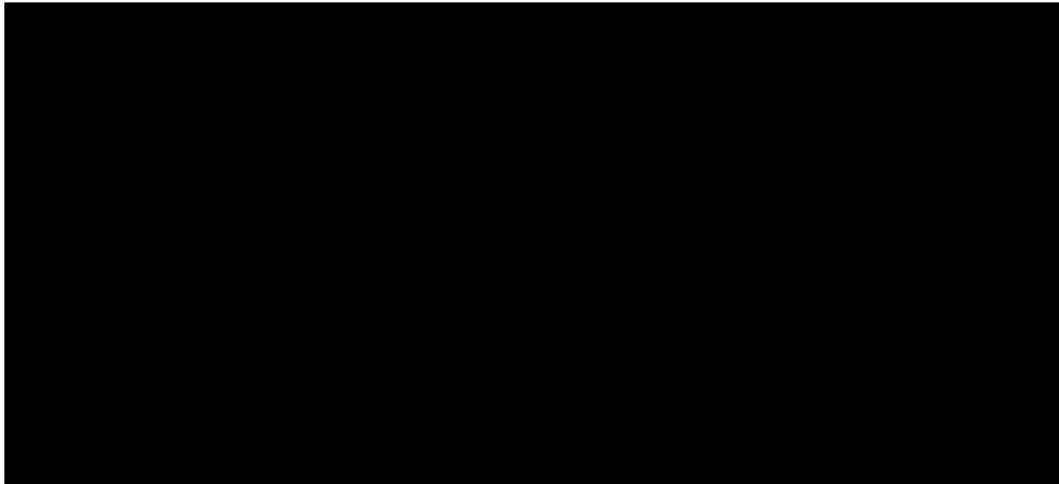
Table 3-4. Joint Base Lewis-McChord condition indices.

Segment	Segment Stations (ft)	Track Structure Condition		Track Component Group Indices		
		Index	Category	RJCI (Rail and Joints)	TCI (Tie)	BSCI (Ballast and Subgrade)
5 Lead	0+00 - 34+50	98	Excellent	98	99	99
Crossover 1	0+00 - 2+11	100	Excellent	100	100	100
Crossover 2	0+00 - 2+12	100	Excellent	100	100	100
Crossover 3	0+00 - 2+08	100	Excellent	100	100	99
Crossover 4	0+00 - 2+09	100	Excellent	100	100	100
Crossover 5	0+00 - 3+17	100	Excellent	100	100	99
Crossover 6	0+00 - 1+69	99	Excellent	100	100	99
Drmo Lead	0+00 - 8+92	99	Excellent	100	99	99
Engine House 1	0+00 - 5+86	100	Excellent	100	100	99
Engine House 2	0+00 - 3+18	100	Excellent	100	100	100
Lc 10	0+00 - 48+04	99	Excellent	99	99	99
Lead 1	0+00 - 36+15	99	Excellent	99	99	99
Lead 2	0+00 - 47+88	99	Excellent	99	99	99
Lead 3	0+00 - 35+27	99	Excellent	100	99	99
Lower 4	0+00 - 16+40	98	Excellent	98	99	99
Madigan Spur	0+00 - 47+21	98	Excellent	100	97	99
Main	0+00 - 52+47	99	Excellent	99	99	99
North 1	0+00 - 23+84	99	Excellent	99	99	99
North 2	0+00 - 26+16	99	Excellent	99	100	99
North 3	0+00 - 29+82	99	Excellent	99	99	100
North Wye	0+00 - 18+63	98	Excellent	99	98	99
South 1	0+00 - 26+13	99	Excellent	99	100	99
South 2	0+00 - 23+13	99	Excellent	99	100	99
South 3	0+00 - 29+83	99	Excellent	99	100	99
South Wye	0+00 - 18+40	98	Excellent	100	98	99
Track 1	0+00 - 30+06	99	Excellent	100	99	99
Track 2	0+00 - 30+99	99	Excellent	99	99	99
Track 3	0+00 - 23+79	100	Excellent	100	100	99
Track 4	0+00 - 31+63	99	Excellent	99	99	99
Track 5	0+00 - 24+11	98	Excellent	97	99	99
Track 6	0+00 - 32+71	99	Excellent	100	99	99
Track 7	0+00 - 24+45	100	Excellent	100	100	99
Track 8	0+00 - 30+96	99	Excellent	100	99	99
Upper 4	0+00 - 11+14	99	Excellent	100	99	99
Upper 6	0+00 - 19+56	99	Excellent	100	99	99

4 Estimate of Maintenance and Repair Costs

[REDACTED]

[illegible]



5 Summary

General summary

This report presents the results of the detailed visual and ultrasonic internal flaw track inspections at Joint Base Lewis-McChord. It provides repair requirements for the track, turnouts, grade crossings, and drainage facilities. Component conditions are given according to the operating restriction and RAILER-computed condition indices. The proposed repairs, if performed, will improve railroad operations and reduce long-term maintenance. Cost estimates are based on estimated national average prices and include materials, equipment, and labor. The repairs indicated are required to upgrade the Joint Base Lewis-McChord railroad facilities to meet the standards as defined in UFC 4-860-03 (Headquarters, Departments of the Army, the Navy, and the Air Force 2008) and to ensure that the track can be maintained at these levels for the foreseeable future with routine maintenance.

Summary of recommendations

Based on this railroad track inspection, the ERDC team recommends the following:

- a. All close-to-traffic defects should be repaired immediately and before any train operations are resumed on those tracks with such defects.
- b. The repairs listed alphabetically in Appendix C by track segment should be performed in the near term.
- c. Continue a program of routine track inspections and preventive maintenance to protect the track-repair investment.
- d. Continue the current crosstie replacement program.
- e. Continue to closely monitor all turnouts and adjust as needed.

References

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_____. 1993b. *Condition index for low volume railroad track: Condition survey inspection and distress manual*. Report FM-93/14. Champaign, IL: U.S. Army Engineer Research and Development Center.

U.S. Army Engineer Research and Development Center. 2008. *RAILER Engineered Management System*.
<http://www.erdc.usace.army.mil/pls/erdcpub/WWW_WELCOME.NAVIGATION_PAGE?tmp_next_page=5384>.

Appendix A: Photos

Examples of defects found during the inspection at Joint Base Lewis-McChord are shown in the following photos.

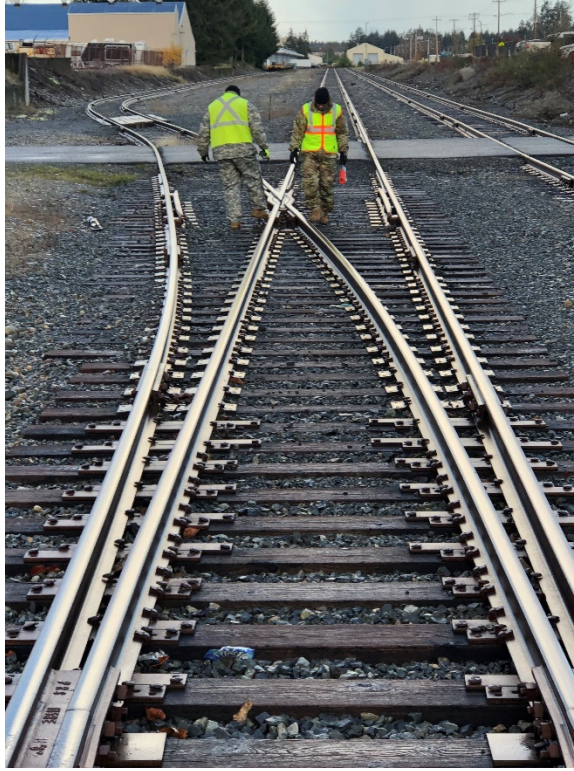


Figure A1. Turnout 37 under evaluation on 5 Lead.



Figure A2. Track 5 Leading into yard tracks.



Figure A3. Storm drain along Lead 3 causing heavy vegetation and potentially causing washouts.



Figure A4. Gate leading to Lead 1, Lead 2 and Lead 3.



Figure A5. View down Main and LC10 toward the North and South tracks.

Appendix B: Inventory of Joint Base Lewis-McChord Railroad Facilities

The inventory of Joint Base Lewis-McChord railroad facilities is presented in this appendix. It includes general track characteristics, turnouts, grade crossings, track appliances, curves, drainage, and bridges.

Table B-1. Joint Base Lewis-McChord General Track Characteristics.

Track	Begin (ft)	End (ft)	Rail Wgt (lb/yd)	Rail Length (ft)	Tie Material	Tie Size (in.)	Joint Bar Length (in.)	Joint Bar, # Holes	Tie Plate, Width (in.)	Tie Plate Length (in.)	Ballast Type
North Wye	0+00	18+63	115	39	Mixed Wood Ties	7"x9"x8.5'	36	6	7	11	#4
South Wye	0+00	18+40	115	39	Mixed Wood Ties	7"x9"x8.5'	36	6	7	11	#4
Crossover 1	0+00	2+11	115	39	Mixed Wood Ties	7"x9"x8.5'	36	6	7	11	#4
Crossover 2	0+00	2+12	115	39	Mixed Wood Ties	7"x9"x8.5'	36	6	7	11	#4
Crossover 3	0+00	2+08	115	39	Mixed Wood Ties	7"x9"x8.5'	36	6	7	11	#4
Crossover 4	0+00	2+09	115	39	Mixed Wood Ties	7"x9"x8.5'	36	6	7	11	#4
Engine House 1	0+00	5+86	115	39	Mixed Wood Ties	7"x9"x8.5'	36	6	7	11	#4
Engine House 2	0+00	3+18	115	39	Mixed Wood Ties	7"x9"x8.5'	24	6	7	11	#4
Lc 10	0+00	48+04	115	39	Mixed Wood Ties	7"x9"x8.5'	36	6	7	11	#4
Madigan Spur	0+00	47+21	115	39	Mixed Wood Ties	7"x9"x8.5'	36	6	7	11	#4
Main	0+00	52+47	115	39	Mixed Wood Ties	7"x9"x8.5'	36	6	7	11	#4
North 1	0+00	23+84	115	39	Mixed Wood Ties	7"x9"x8.5'	36	6	7	11	#4
North 2	0+00	26+16	115	39	Mixed Wood Ties	7"x9"x8.5'	36	6	7	11	#4
North 3	0+00	29+82	115	39	Mixed Wood Ties	7"x9"x8.5'	36	6	7	11	#4
South 1	0+00	26+13	115	39	Mixed Wood Ties	7"x9"x8.5'	24	6	7	11	#4
South 2	0+00	23+13	115	39	Mixed Wood Ties	7"x9"x8.5'	36	6	7	11	#4
South 3	0+00	29+83	115	39	Mixed Wood Ties	7"x9"x8.5'	36	6	7	11	#4
Track 1	0+00	30+06	115	39	Mixed Wood Ties	7"x9"x8.5'	36	6	7	11	#4
Track 2	0+00	30+99	115	39	Mixed Wood Ties	7"x9"x8.5'	36	6	7	11	#4
Track 3	0+00	23+79	115	39	Mixed Wood Ties	7"x9"x8.5'	36	6	7	11	#4
Track 4	0+00	31+63	115	39	Mixed Wood Ties	7"x9"x8.5'	36	6	7	11	#4
Track 5	0+00	24+11	115	39	Mixed Wood Ties	7"x9"x8.5'	24	6	7	11	#4
Track 6	0+00	32+71	115	39	Mixed Wood Ties	7"x9"x8.5'	24	6	7	11	#4
Track 7	0+00	24+45	115	39	Mixed Wood Ties	7"x9"x8.5'	24	6	7	11	#4
Track 8	0+00	30+96	115	39	Mixed Wood Ties	7"x9"x8.5'	24	6	7	11	#4
Upper 4	0+00	11+14	115	39	Mixed Wood Ties	7"x9"x8.5'	24	6	7	11	#4
Upper 6	0+00	19+56	115	39	Mixed Wood Ties	7"x9"x8.5'	24	6	7	11	#4
5 Lead	0+00	34+50	115	39	Mixed Wood Ties	7"x9"x8.5'	36	6	7	11	#4
Crossover 5	0+00	3+17	115	39	Mixed Wood Ties	7"x9"x8.5'	24	6	7	11	#4
Crossover 6	0+00	1+69	115	39	Mixed Wood Ties	7"x9"x8.5'	24	6	7	11	#4
Drmo Lead	0+00	8+92	115	39	Mixed Wood Ties	7"x9"x8.5'	24	6	7	11	#4
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Track	Begin (ft)	End (ft)	Rail Wgt (lb/yd)	Rail Length (ft)	Tie Material	Tie Size (in.)	Joint Bar Length (in.)	Joint Bar, # Holes	Tie Plate, Width (in.)	Tie Plate Length (in.)	Ballast Type
Lead 1	0+00	36+15	115	39	Mixed Wood Ties	7"x9"x8.5'	24	6	7	11	#4
Lead 2	0+00	47+88	115	39	Mixed Wood Ties	7"x9"x8.5'	36	6	7	11	#4
Lead 3	0+00	35+27	115	39	Mixed Wood Ties	7"x9"x8.5'	24	6	7	11	#4
Lower 4	0+00	16+40	115	39	Mixed Wood Ties	7"x9"x8.5'	24	6	7	11	#4

Table B-2. Joint Base Lewis-McChord Turnouts.

Track	Turnout Name	Switch Location (ft)	Orientation	Direction	Frog Type	Frog Size	Comments
5 Lead	35	6+61	Facing Point	Right Hand	Solid Self Guarded Manganese	9	
5 Lead	37	18+13	Facing Point	Left Hand	Rail Bound Manganese	10	
5 Lead	36	14+96	Facing Point	Right Hand	Rail Bound Manganese	10	
Engine House 1	29	2+56	Facing Point	Right Hand	Rail Bound Manganese	8	
Lc 10	6	8+37	Facing Point	Right Hand	Rail Bound Manganese	10	
Lc 10	4	7+61	Trailing Point	Right Hand	Solid Self Guarded Manganese	10	
Lc 10	11	16+86	Facing Point	Left Hand	Solid Self Guarded Manganese	10	
Lc 10	20	38+13	Trailing Point	Left Hand	Rail Bound Manganese	10	
Lc 10	22	40+44	Trailing Point	Right Hand	Solid Self Guarded Manganese	10	
Lc 10	14	25+13	Trailing Point	Right Hand	Solid Self Guarded Manganese	10	
Lc 10	23	40+83	Facing Point	Right Hand	Solid Self Guarded Manganese	10	
Lc 10	25	42+44	Facing Point	Right Hand	Solid Self Guarded Manganese	10	
Lc 10	28	45+01	Facing Point	Right Hand	Solid Self Guarded Manganese	8	
Lc 10	27	43+57	Facing Point	Right Hand	Rail Bound Manganese	10	
Lead 1	43	14+66	Facing Point	Right Hand	Solid Self Guarded Manganese	8	
Lead 2	41	11+60	Facing Point	Left Hand	Solid Self Guarded Manganese	8	
Lead 2	45	30+55	Facing Point	Right Hand	Solid Self Guarded Manganese	8	
Lead 2	40	10+54	Trailing Point	Left Hand	Solid Self Guarded Manganese	8	
Lead 2	44	29+40	Trailing Point	Right Hand	Bolted Rigid	8	
Lead 2	42	12+60	Facing Point	Right Hand	Solid Self Guarded Manganese	8	
Lead 3	46	19+65	Trailing Point	Right Hand	Solid Self Guarded Manganese	8	
Madigan Spur	38	0+00	Facing Point	Right Hand	Solid Self Guarded Manganese	8	
Madigan Spur	39	9+01	Trailing Point	Equilateral	Solid Self Guarded Manganese	8	heel bolt loose on left
Main	19	35+79	Trailing Point	Right Hand	Rail Bound Manganese	10	
Main	5	6+06	Facing Point	Left Hand	Rail Bound Manganese	10	
Main	1	52+47	Facing Point	Right Hand	Solid Self Guarded Manganese	10	3 bolts in heel filler loose
Main	2	0+00	Trailing Point	Right Hand	Solid Self Guarded Manganese	10	
Main	3	3+20	Facing Point	Right Hand	Solid Self Guarded Manganese	10	
Main	12	16+64	Trailing Point	Left Hand	Solid Self Guarded Manganese	10	
Main	33	45+69	Trailing Point	Left Hand	Solid Self Guarded Manganese	10	
Main	21	36+01	Facing Point	Right Hand	Solid Self Guarded Manganese	10	
Main	13	20+69	Facing Point	Right Hand	Solid Self Guarded Manganese	10	

Continued on next page.

Track	Turnout Name	Switch Location (ft)	Orientation	Direction	Frog Type	Frog Size	Comments
Main	34	46+12	Facing Point	Right Hand	Rail Bound Manganese	10	
Main	26	40+69	Facing Point	Left Hand	Rail Bound Manganese	10	
North 2	16	25+00	Trailing Point	Right Hand	Rail Bound Manganese	10	
North 2	9	1+15	Facing Point	Left Hand	Rail Bound Manganese	10	
North 3	17	28+01	Trailing Point	Right Hand	Rail Bound Manganese	10	
North 3	8	1+82	Facing Point	Left Hand	Rail Bound Manganese	10	
South 3	7	1+82	Facing Point	Right Hand	Rail Bound Manganese	10	
South 3	10	3+31	Facing Point	Right Hand	Rail Bound Manganese	10	
South 3	18	28+02	Trailing Point	Left Hand	Solid Self Guarded Manganese	10	
South 3	15	26+49	Trailing Point	Left Hand	Bolted Rigid	10	
Track 2	32	7+18	Facing Point	Right Hand	Solid Self Guarded Manganese	10	
Track 4	31	7+51	Facing Point	Right Hand	Solid Self Guarded Manganese	10	
Track 6	30	8+27	Facing Point	Right Hand	Solid Self Guarded Manganese	10	
Track 6	24	1+16	Facing Point	Right Hand	Solid Self Guarded Manganese	10	

Table B-3. Joint Base Lewis-McChord Grade Crossings.

Track	Road Name	Begin Station (ft)	End Station (ft)	Length (ft)	Material	Flangeway	Comments
5 Lead	Ranier St	5+31.73	6+12.93	81.2	Asphalt	Asphalt	
5 Lead	Service Rd	9+73.34	9+92.57	19.23	Asphalt	Asphalt	
5 Lead	South O	19+21.37	19+51.82	30.45	Asphalt	Asphalt	
LC 10	Lincoln St	7+81.98	8+19.80	37.82	Concrete (precast panel)	Rubber	
Lead 1	S I street	21+11.34	21+47.09	35.75	Concrete (precast panel)	Rubber	
Lead 2	South L St	32+70.37	33+6.74	36.37	Concrete (precast panel)	Rubber	
Lead 2	bay c crossing	42+72.75	43+2.88	30.13	Asphalt	Asphalt	
Lead 2	bay b crossing	43+54.92	43+73.20	18.28	Asphalt	Asphalt	
Lead 2	bay a crossing	45+96.17	46+13.98	17.81	Asphalt	Asphalt	
Lead 3	bay b crossing	30+96.62	31+13.84	17.22	Asphalt	Asphalt	
Lead 3	S I street	20+7.76	20+45.25	37.49	Concrete (precast panel)	Rubber	
Lead 3	bay c	30+14.53	30+44.96	30.43	Asphalt	Asphalt	
Lead 3	bay a cossing	33+36.58	33+55.19	18.61	Asphalt	Asphalt	
Lower 4	south o st.	1+9.79	1+33.78	23.99	Asphalt	Asphalt	
Madigan Spur	madigan crossing 3	11+62.21	11+90.50	28.29	Asphalt	Asphalt	
Madigan Spur	madigan crossing 2	6+64.84	6+92.14	27.3	Asphalt	Asphalt	
Madigan Spur	dog kennel crossing	42+21.62	42+51.01	29.39	Asphalt	Asphalt	
Madigan Spur	boiler plant crossin	38+17.69	38+45.43	27.74	Asphalt	Asphalt	
Madigan Spur	madigan crossing	3+78.03	4+4.32	26.29	Asphalt	Asphalt	
Main	Ranier Dr	50+44.83	50+88.59	43.76	Concrete (precast panel)	Rubber	
Main	Lincoln St	5+45.96	5+81.17	35.21	Concrete (precast panel)	Rubber	
North 2	timber crossing	0+95.44	1+12.90	17.46	Timber (wood)	None	
North 2	timber crossing	25+10.87	25+28.39	17.52	Timber (wood)	None	
North 3	maint	2+77.13	2+94.53	17.4	Timber (wood)	Formed Steel	
North Wye	maintenance	1+40.01	1+68.48	28.47	Asphalt	Asphalt	
South 2	timber crossing	21+66.13	21+85.38	19.25	Timber (wood)	None	
South 2	timber crossing	1+48.71	1+67.28	18.57	Timber (wood)	None	
South 3	timber crossing	3+2.30	3+20.53	18.23	Timber (wood)	None	
South 3	timber crossing	26+58.33	26+79.48	21.15	Timber (wood)	None	
South 3	timber crossing	4+81.66	5+0.22	18.56	Timber (wood)	None	
South 3	timber crossing	25+2.22	25+22.44	20.22	Timber (wood)	None	
South Wye	maintenance	2+24.44	2+79.09	54.65	Asphalt	Asphalt	
Track 2	maintenance	4+81.57	4+93.75	12.18	Timber (wood)	None	

Track	Road Name	Begin Station (ft)	End Station (ft)	Length (ft)	Material	Flangeway	Comments
Track 4	maintenance	5+29.93	5+42.48	12.55	Timber (wood)	None	
Track 6	maintenance	7+43.99	7+59.16	15.17	Timber (wood)	None	
Track 8	yard crossing	9+54.17	9+80.22	26.05	Concrete (precast panel)	Concrete	
Track 8	maintenance	2+69.40	3+8.46	39.06	Asphalt	Asphalt	
Upper 4	maintenance	2+84.73	3+63.62	78.89	Asphalt	Asphalt	

Table B-4. Joint Base Lewis-McChord Track Appliances.

Track	Station (ft)	Appliance	Comments
Lead 1	36+15	Car Bumper	
Upper 4	11+14	Car Bumper	
Madigan Spur	47+21	Car Bumper	
Madigan Spur	12+53	Gate	
South 3	1+70	Derail	
Main	51+68	Gate	
Main	38+14	Gate	
Main	6+31	Gate	
Lc 10	0+83	Gate	
Track 8	30+96	End Ramp (Concrete)	
Track 3	23+79	End Ramp (Concrete)	
Track 2	30+99	End Ramp (Concrete)	
Track 1	30+06	Wheel Stop	
Track 1	8+45	Gate	
South Wye	16+71	Gate	
North 3	28+14	Derail	
Track 5	24+11	End Ramp (Concrete)	
Track 6	32+71	End Ramp (Concrete)	
Track 6	11+09	Gate	
Track 5	2+46	Gate	
Track 4	10+04	Gate	
Track 7	2+80	Gate	
Track 7	24+45	End Ramp (Concrete)	
Track 4	31+63	End Ramp (Concrete)	
Engine House 1	5+86	Door	
Engine House 2	3+18	Door	
5 Lead	34+50	Car Bumper	
Upper 6	19+56	Car Bumper	
Lower 4	16+40	Car Bumper	
Madigan Spur	14+40	Derail	
Madigan Spur	3+11	Gate	
Drmo Lead	6+09	Gate	
Continued on next page.			

Track	Station (ft)	Appliance	Comments
Lead 1	1+20	Gate	
Lead 2	47+88	End Ramp (Concrete)	
Lead 3	35+27	End Ramp (Concrete)	
Lead 3	0+19	Gate	
Lead 2	7+69	Gate	
North 3	1+72	Derail	
Engine House 2	1+38	Derail	
North Wye	16+93	Gate	
Track 1	7+62	Derail	
Engine House 1	3+98	Derail	
Track 6	10+15	Derail	
Track 8	3+89	Derail	
Lead 2	12+80	Gate	
Lead 2	32+15	Derail	
South 3	28+12	Derail	
Track 5	1+82	Derail	
Track 4	9+39	Derail	
Track 2	9+38	Gate	
Lead 3	19+71	Derail	
Track 2	8+95	Derail	
Track 7	1+92	Derail	
Track 3	1+74	Derail	

Table B-5. Joint Base Lewis-McChord Bridges.

Track	Bridge ID	Begin Station (ft)	End Station (ft)	Length (ft)	Const Type	Ballasted?	Comments
No data available.							

Table B-6. Joint Base Lewis-McChord Curves.

Track	Begin Station (ft)	End Station (ft)	Length (ft)	Direction	Curvature (deg)	Superelevation (in.)	Comments
South Wye	0+01	16+71	1669.99	Left	5	0	
North Wye	6+85	18+63	1178.38	Right	4	0	
Track 8	0+11	6+43	631.75	Right	5	0	
Track 6	5+51	11+07	555.65	Right	5	0	
Track 4	0+03	10+01	998.24	Right	4	0	
Track 1	0+20	6+61	640.9	Right	4	0	
Track 2	0+10	8+83	872.74	Right	4	0	
5 Lead	0+02	10+88	1086.23	Right	4	0	
Lead 2	0+54	15+60	1505.97	Right	4	0	
Madigan Spur	34+77	39+29	452.14	Right	3	0	
Madigan Spur	28+50	33+60	509.76	Right	3	0	
Madigan Spur	9+09	16+40	730.84	Right	4	0	
Madigan Spur	2+24	8+95	670.63	Left	4	0	
Ormo Lead	2+07	6+08	401.33	Left	5	1	

Table B-7. Joint Base Lewis-McChord Drainage.

Track	Station (ft)	Drainage Type	Height (in.)	Width (in.)	Diameter (in.)	Length (ft)	Drainage Material	Comments
Track 5	23+98	Drainage Structure	0	0	24	0	CMP	
Track 7	17+79	Drainage Structure	0	0	24	0	CMP	
Track 1	16+76	Drainage Structure	0	0	24	0	CMP	
Track 7	11+07	Drainage Structure	0	0	24	0	CMP	
Track 6	32+57	Drainage Structure	0	0	24	0	CMP	
Track 1	30+06	Drainage Structure	0	0	24	0	CMP	
Track 3	17+10	Drainage Structure	0	0	24	0	CMP	
Track 5	17+45	Drainage Structure	0	0	24	0	CMP	
Track 3	23+76	Drainage Structure	0	0	24	0	CMP	
Track 6	19+35	Drainage Structure	0	0	24	0	CMP	
Track 4	31+51	Drainage Structure	0	0	24	0	CMP	

Appendix C: Defects and Recommended Repair

Defects, repair recommendations, and cost estimates for Joint Base Lewis-McChord are shown in this appendix in alphabetical order by segment name.

Each table consists of the list of defects for one segment. At the top of the table, general information, including the area, track, begin location, end location, and track category, for the segment is provided. Defects are listed by their component. The components include the ties, rail, fastenings and other track materials (F&OTM), ballast/subgrade, drainage, geometry, appliances, turnouts, and grade crossings. The total cost to repair the segment to a defect-free standard is shown at the end of each table.

Each defect is described further to assist in locating and repairing the defect. The “Station (ft)” column is the location of the defect. If defects are found repeatedly for a segment, the individual stations are not provided; instead, the station will reflect the length of the segment, and the defect quantity will be provided. The “Rail” column refers to which side (right, left, or both) of the track the defect was found. The “Standard” column refers to the maintenance and safety standards listed in UFC 4-860-03. The “Work Action” column lists the recommended plan of action for repairing the defect found. Measurements for the defect (i.e., track gauge or switch point gap) are listed in the “Meas (in.)” column. The quantity of the defect found (i.e., number of defective crossties) is listed in the “Quant” column. The unit of measure for the defect is listed in the “UM” column. Units of measure include the following: each (EA), linear foot (LF), and track foot (TF). The unit cost is listed in the “Unit Cost” column. The total/extended cost is the unit cost multiplied by the quantity and is represented in the “Ext. Cost” column. Any notes for the defects are listed in the last column.

5 Lead								
Area: Main Post			Segment Begin Location: 0+0.00			Track Category: B		
Track: 5 Lead			Segment End Location: 34+50.20					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)
Ties								
Single Defective Tie	0+00 - 34+50	N/A	Maint A	Install/Replace One Tie (7"x9"x8.5')		14	EA	
Single Missing Tie	0+00 - 34+50	N/A	Maint A	Install/Replace One Tie (7"x9"x8.5')		1	EA	
Rail								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
OTM								
Bolts, Loose (1 Bolt) & Broken/Missing Washer	0+00	B	Maint B	Install or Replace One Bolt		5	EA	
Rail End Mis (Gage) > 0.125" & <= 0.1875" (>1/8 - <=3/16)	28+02	R	Maint B	Eliminate Rail Mismatch		1	EA	
Ballast/Subgrade								
Dirty (Fouled) Ballast	3+82 - 34+45	B	Maint A	Clean Ballast (Undercut)		3063	TF	
Drainage								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Geometry								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Appliances								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Turnout: 35								
Surface and Alignment (Good)	6+61 - 7+67	N/A	Maint A	Do Nothing		1		
Rail Braces - Rigid (Left), Loose	6+61 - 7+67	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		1	EA	
Rail Braces - Rigid (Right), Loose	6+61 - 7+67	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		4	EA	
Gauge At Switch Points	6+61 - 7+67	N/A	Full Compl	Correct Gauge	56.75	1	EA	
Turnout: 36								
Rail Braces - Rigid (Left), Loose	14+96 - 16+12	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		4	EA	
Continued on next page.								

5 Lead									
Area: Main Post			Segment Begin Location: 0+0.00				Track Category: B		
Track: 5 Lead			Segment End Location: 34+50.20						
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM		Note(s)
Surface and Alignment (Good)	14+96 - 16+12	N/A	Maint A	Do Nothing		1			
Rail Braces - Rigid (Right), Loose	14+96 - 16+12	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		1	EA		
Gauge At Joints In Curved Closure Rail (2nd)	14+96 - 16+12	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.625	1			
Gauge At Joints In Curved Closure Rail (1st)	14+96 - 16+12	N/A	Full Compl	Correct Gauge	56.375	1	EA		
Switch Ties, Defective (12 ft)	14+96 - 16+12	N/A	Maint A	Install/Replace 12' Switch Tie		3	EA		
Turnout: 37									
Rail Braces - Rigid (Left), Loose	18+13 - 19+29	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		1	EA		
Gauge At Joints In Curved Closure Rail (2nd)	18+13 - 19+29	N/A	Full Compl	Correct Gauge	56.75	1	EA		
Rail Braces - Rigid (Right), Loose	18+13 - 19+29	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		3	EA		
Surface and Alignment (Good)	18+13 - 19+29	N/A	Maint A	Do Nothing		1			
Gauge At Frog Point (Right)	18+13 - 19+29	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.75	1			
Grade Crossings									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		A
Total Cost									

Crossover 3								
Area: Logistics			Segment Begin Location: 0+0.00			Track Category: B		
Track: Crossover 3			Segment End Location: 2+8.32					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)
Ties								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Rail								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
OTM								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Ballast/Subgrade								
Dirty (Fouled) Ballast	0+03 - 2+08	B	Maint A	Clean Ballast (Undercut)		205	TF	
Drainage								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Geometry								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Appliances								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Turnouts								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Grade Crossings								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Total Cost								

Crossover 5								
Area: Main Post			Segment Begin Location: 0+0.00			Track Category: A		
Track: Crossover 5			Segment End Location: 3+16.76					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)
Ties								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Rail								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
OTM								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Ballast/Subgrade								
Dirty (Fouled) Ballast	0+65 - 3+15	B	Maint A	Clean Ballast (Undercut)		250	TF	
Drainage								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Geometry								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Appliances								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Turnouts								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Grade Crossings								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Total Cost								

Crossover 6									
Area: Main Post				Segment Begin Location: 0+0.00				Track Category: A	
Track: Crossover 6				Segment End Location: 1+69.39					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM		Note(s)
Ties									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		
Rail									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		
OTM									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		
Ballast/Subgrade									
Dirty (Fouled) Ballast	0+01 - 1+69	B	Maint A	Clean Ballast (Undercut)		168	TF		
Drainage									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		
Geometry									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		
Appliances									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		
Turnouts									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		
Grade Crossings									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		
Total Cost									

Drmo lead									
Area: Main Post				Segment Begin Location: 0+0.00				Track Category: A	
Track: Drmo lead				Segment End Location: 8+92.12					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM		Note(s)
Ties									
Single Defective Tie	0+00 - 8+92	N/A	Maint A	Install/Replace One Tie (7"x9"x8.5')		2	EA		
Rail									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		
OTM									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		
Ballast/Subgrade									
Dirty (Fouled) Ballast	2+35 - 8+92	B	Maint A	Clean Ballast (Undercut)		657	TF		
Drainage									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		
Geometry									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		
Appliances									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		
Turnouts									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		
Grade Crossings									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		
Total Cost									

Engine House 1								
Area: Logistics			Segment Begin Location: 0+0.00			Track Category: B		
Track: Engine House 1			Segment End Location: 5+85.71					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)
Ties								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	/A
Rail								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	/A
OTM								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	/A
Ballast/Subgrade								
Dirty (Fouled) Ballast	5+86	B	Maint A	Clean Ballast (Undercut)		1	TF	
Drainage								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	/A
Geometry								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	/A
Appliances								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	/A
Turnout: 29								
Gauge At Joints In Curved Closure Rail (2nd)	2+56 - 3+54	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.625	1		
Gauge At Joints In Curved Closure Rail (1st)	2+56 - 3+54	N/A	Full Compl	Correct Gauge	56.625	1	EA	
Rail Braces - Rigid (Right), Loose	2+56 - 3+54	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		4	EA	
Rail Braces - Rigid (Left), Loose	2+56 - 3+54	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		1	EA	
Switch Ties, Defective (16 ft)	2+56 - 3+54	N/A	Maint A	Install/Replace 16' Switch Tie		1	EA	
Surface and Alignment (Good)	2+56 - 3+54	N/A	Maint A	Do Nothing		1		
Grade Crossings								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	/A
Total Cost								

LC 10								
Area: Logistics			Segment Begin Location: 0+0.00			Track Category: B		
Track: LC 10			Segment End Location: 48+3.99					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)
Ties								
Single Defective Tie	0+00 - 48+04	N/A	Maint A	Install/Replace One Tie (7"x9"x8.5')		22	EA	
Rail								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A
OTM								
Rail End Mis (Tread) > 0.125" & <= 0.1875" (>1/8 - <=3/16)	20+96	R	Maint B	Eliminate Rail Mismatch		1	EA	
Bolts, Loose (1 Bolt) & Broken/Missing Washer	0+00	B	Maint B	Install or Replace One Bolt		8	EA	
Ballast/Subgrade								
Dirty (Fouled) Ballast	4+45 - 47+99	B	Maint A	Clean Ballast (Undercut)		4354	TF	
Drainage								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A
Geometry								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A
Appliances								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A
Turnout: 11								
Surface and Alignment (Good)	16+86 - 18+02	N/A	Maint A	Do Nothing		1		
Switch Ties, Defective (16 ft)	16+86 - 18+02	N/A	Maint A	Install/Replace 16' Switch Tie		1	EA	
Rail Braces - Rigid (Left), Loose	16+86 - 18+02	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		2	EA	
Rail Braces - Rigid (Right), Loose	16+86 - 18+02	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		1	EA	
Switch Rods, Loose	16+86 - 18+02	N/A	Close to Traffic	Tighten/Adjust Switch Rod		1	EA	
Gauge At Joints In Curved Closure Rail (2nd)	16+86 - 18+02	N/A	Full Compl	Correct Gauge	56.75	1	EA	
Turnout: 14								
Continued on next page.								

LC 10								
Area: Logistics			Segment Begin Location: 0+0.00				Track Category: B	
Track: LC 10			Segment End Location: 48+3.99					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)
Surface and Alignment (Good)	23+97 - 25+13	N/A	Maint A	Do Nothing		1		
Rail Braces - Rigid (Right), Loose	23+97 - 25+13	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		5	EA	
Turnout: 20								
Gauge At Joints In Curved Closure Rail (1st)	36+97 - 38+13	N/A	Full Compl	Correct Gauge	56.25	1	EA	
Gauge At Joints In Curved Closure Rail (2nd)	36+97 - 38+13	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.75	1		
Gauge At Frog Point (Right)	36+97 - 38+13	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.25	1		
Surface and Alignment (Good)	36+97 - 38+13	N/A	Maint A	Do Nothing		1		
Rail Braces - Rigid (Right), Loose	36+97 - 38+13	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		3	EA	
Turnout: 22								
Surface and Alignment (Good)	39+28 - 40+44	N/A	Maint A	Do Nothing		1		
Rail Braces - Rigid (Left), Loose	39+28 - 40+44	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		6	EA	
Rail Braces - Rigid (Right), Loose	39+28 - 40+44	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		5	EA	
Frog Bolts, Loose	39+28 - 40+44	N/A	Maint A	Tighten Frog Bolt		1	EA	
Gauge At Frog Point (Left)	39+28 - 40+44	N/A	Full Compl	Correct Gauge	56.25	1	EA	
Turnout: 23								
Gauge At Joints In Curved Closure Rail (2nd)	40+83 - 41+99	N/A	Full Compl	Correct Gauge	56.75	1	EA	
Frog Flangeway Depth (Left)	40+83 - 41+99	N/A	Full Compl	Rebuild Frog	1.75	1	EA	
Gauge At Frog Point (Left)	40+83 - 41+99	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.875	1		
Frog Flangeway Depth (Right)	40+83 - 41+99	N/A	Full Compl	Rebuild Frog	1.75	1	EA	
Surface and Alignment (Good)	40+83 - 41+99	N/A	Maint A	Do Nothing		1		
Continued on next page.								

LC 10								
Area: Logistics			Segment Begin Location: 0+0.00				Track Category: B	
Track: LC 10			Segment End Location: 48+3.99					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)
Rail Braces - Rigid (Left), Loose	40+83 - 41+99	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		4	EA	
Rail Braces - Rigid (Right), Loose	40+83 - 41+99	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		5	EA	
Gauge At Frog Point (Right)	40+83 - 41+99	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.875	1		
Turnout: 25								
Rail Braces - Rigid (Right), Loose	42+44 - 43+60	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		7	EA	
Gauge At Joints In Curved Closure Rail (1st)	42+44 - 43+60	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.75	1		
Gauge At Joints In Curved Closure Rail (2nd)	42+44 - 43+60	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.75	1		
Gauge At Switch Points	42+44 - 43+60	N/A	Full Compl	Correct Gauge	56.25	1	EA	
Surface and Alignment (Good)	42+44 - 43+60	N/A	Maint A	Do Nothing		1		
Rail Braces - Rigid (Left), Loose	42+44 - 43+60	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		7	EA	
Turnout: 27								
Surface and Alignment (Good)	43+57 - 44+73	N/A	Maint A	Do Nothing		1		
Rail Braces - Rigid (Right), Loose	43+57 - 44+73	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		4	EA	
Rail Braces - Rigid (Left), Loose	43+57 - 44+73	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		4	EA	
Switch Rods, Loose	43+57 - 44+73	N/A	Close to Traffic	Tighten/Adjust Switch Rod		1	EA	
Gauge At Frog Point (Right)	43+57 - 44+73	N/A	Full Compl	Correct Gauge	56.375	1	EA	
Turnout: 28								
Rail Braces - Rigid (Left), Loose	45+01 - 45+99	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		2	EA	
Gauge At Frog Point (Left)	45+01 - 45+99	N/A	Full Compl	Correct Gauge	56.375	1	EA	
Rail Braces - Rigid (Right), Loose	45+01 - 45+99	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		5	EA	
Surface and Alignment (Good)	45+01 - 45+99	N/A	Maint A	Do Nothing		1		

LC 10								
Area: Logistics			Segment Begin Location: 0+0.00				Track Category: B	
Track: LC 10			Segment End Location: 48+3.99					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)
Turnout: 4								
Rail Braces - Rigid (Left), Loose	6+45 - 7+61	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		4	EA	
Surface and Alignment (Good)	6+45 - 7+61	N/A	Maint A	Do Nothing		1		
Switch Rods, Loose	6+45 - 7+61	N/A	Close to Traffic	Tighten/Adjust Switch Rod		1	EA	
Switch Ties, Defective (10 ft)	6+45 - 7+61	N/A	Maint A	Install/Replace 10' Switch Tie		1	EA	
Turnout: 6								
Surface and Alignment (Good)	8+37 - 9+53	N/A	Maint A	Do Nothing		1		
Rail Braces - Rigid (Left), Loose	8+37 - 9+53	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		3	EA	
Rail Braces - Rigid (Right), Loose	8+37 - 9+53	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		5	EA	
Gauge At Frog Point (Left)	8+37 - 9+53	N/A	Full Compl	Correct Gauge	56.625	1	EA	
Gauge At Frog Point (Right)	8+37 - 9+53	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.375	1		
Guard Rail Flangeway Width (Left)	8+37 - 9+53	N/A	Full Compl	Tighten/Adjust Guard Rail Filler (Separator Block)	1.75	1	EA	
Grade Crossings								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Total Cost								

Lead 1									
Area: Main Post				Segment Begin Location: 0+0.00				Track Category: A	
Track: Lead 1				Segment End Location: 36+15.18					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)	
Ties									
Single Defective Tie	0+00 - 36+15	N/A	Maint A	Install/Replace One Tie (7"x9"x8.5')		11	EA		
Rail									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A	
OTM									
Bolts, Loose (1 Bolt) & Broken/Missing Washer	0+00	B	Maint B	Install or Replace One Bolt		2	EA		
Ballast/Subgrade									
Dirty (Fouled) Ballast	1+02 - 36+07	B	Maint A	Clean Ballast (Undercut)		3505	TF		
Drainage									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A	
Geometry									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A	
Appliances									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A	
Turnouts									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A	
Grade Crossings									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A	
Total Cost									

Lead 2								
Area: Main Post			Segment Begin Location: 0+0.00			Track Category: B		
Track: Lead 2			Segment End Location: 47+87.58					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)
Ties								
Single Defective Tie	0+00 - 47+88	N/A	Maint A	Install/Replace One Tie (7"x9"x8.5')		10	EA	
Rail								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
OTM								
Bolts, Loose (1 Bolt) & Broken/Missing Washer	0+00	B	Maint B	Install or Replace One Bolt		2	EA	
Gauge Rod, Loose	10+63	B	Maint A	Tighten Gauge Rod(s)		1	EA	
Ballast/Subgrade								
Dirty (Fouled) Ballast	16+64 - 29+38	B	Maint A	Clean Ballast (Undercut)		1274	TF	
Dirty (Fouled) Ballast	29+39 - 47+88	B	Maint A	Clean Ballast (Undercut)		1848	TF	
Dirty (Fouled) Ballast	1+08 - 16+62	B	Maint A	Clean Ballast (Undercut)		1554	TF	
Drainage								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Geometry								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Appliances								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Turnout: 40								
Surface and Alignment (Good)	9+56 - 10+54	N/A	Maint A	Do Nothing		1		
Gauge At Joints In Curved Closure Rail (2nd)	9+56 - 10+54	N/A	Full Compl	Correct Gauge	56.75	1	EA	
Rail Braces - Rigid (Left), Loose	9+56 - 10+54	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		1	EA	
Rail Braces - Rigid (Right), Loose	9+56 - 10+54	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		2	EA	
Turnout: 41								
Gauge At Joints In Curved Closure Rail (2nd)	11+60 - 12+58	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.625	1		
Continued on next page.								

Lead 2								
Area: Main Post			Segment Begin Location: 0+0.00				Track Category: B	
Track: Lead 2			Segment End Location: 47+87.58					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)
Gauge At Joints In Curved Closure Rail (1st)	11+60 - 12+58	N/A	Full Compl	Correct Gauge	56.625	1	EA	
Rail Braces - Rigid (Right), Loose	11+60 - 12+58	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		5	EA	
Rail Braces - Rigid (Left), Loose	11+60 - 12+58	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		4	EA	
Switch Ties, Defective (16 ft)	11+60 - 12+58	N/A	Maint A	Install/Replace 16' Switch Tie		1	EA	
Surface and Alignment (Good)	11+60 - 12+58	N/A	Maint A	Do Nothing		1		
Turnout: 42								
Surface and Alignment (Good)	12+60 - 13+58	N/A	Maint A	Do Nothing		1		
Rail Braces - Rigid (Right), Loose	12+60 - 13+58	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		3	EA	
Rail Braces - Rigid (Left), Loose	12+60 - 13+58	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		3	EA	
Frog Flangeway Depth (Right)	12+60 - 13+58	N/A	Full Compl	Rebuild Frog	1.75	1	EA	
Gauge At Frog Point (Left)	12+60 - 13+58	N/A	Full Compl	Correct Gauge	56.625	1	EA	
Frog Flangeway Depth (Left)	12+60 - 13+58	N/A	Full Compl	Rebuild Frog	1.75	1	EA	
Turnout: 44								
Surface and Alignment (Good)	28+42 - 29+40	N/A	Maint A	Do Nothing		1		
Rail Braces - Rigid (Right), Loose	28+42 - 29+40	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		3	EA	
Rail Braces - Rigid (Left), Loose	28+42 - 29+40	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		2	EA	
Frog Flangeway Depth (Right)	28+42 - 29+40	N/A	Full Compl	Rebuild Frog	1.75	1	EA	
Frog Flangeway Depth (Left)	28+42 - 29+40	N/A	Full Compl	Rebuild Frog	1.75	1	EA	
Gauge At Joints In Curved Closure Rail (1st)	28+42 - 29+40	N/A	Full Compl	Correct Gauge	56.375	1	EA	
Turnout: 45								
Surface and Alignment (Good)	30+55 - 31+53	N/A	Maint A	Do Nothing		1		

Lead 2									
Area: Main Post				Segment Begin Location: 0+0.00				Track Category: B	
Track: Lead 2				Segment End Location: 47+87.58					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)	
Switch Ties, Defective (10 ft)	30+55 - 31+53	N/A	Maint A	Install/Replace 10' Switch Tie		1	EA		
Frog Bolts, Improper Size	30+55 - 31+53	N/A	Maint A	Install or Replace Frog Bolt Assembly		1	EA		
Switch Rods, Loose	30+55 - 31+53	N/A	Close to Traffic	Tighten/Adjust Switch Rod		1	EA		
Rail Braces - Rigid (Right), Loose	30+55 - 31+53	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		2	EA		
Point Rails (Right), Improper Type	30+55 - 31+53	N/A	Close to Traffic	Install or Replace Point Rail		1	EA		
Rail Braces - Rigid (Left), Loose	30+55 - 31+53	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		4	EA		
Gauge At Frog Point (Left)	30+55 - 31+53	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.375	1			
Gauge At Joints In Curved Closure Rail (2nd)	30+55 - 31+53	N/A	Full Compl	Correct Gauge	56.625	1	EA		
Gauge At Frog Point (Right)	30+55 - 31+53	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.375	1			
Grade Crossings									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A	
Total Cost									

Lead 3								
Area: Main Post			Segment Begin Location: 0+0.00			Track Category: A		
Track: Lead 3			Segment End Location: 35+27.27					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)
Ties								
Single Defective Tie	0+00 - 35+27	N/A	Maint A	Install/Replace One Tie (7"x9"x8.5')		5	EA	
Rail								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
OTM								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Ballast/Subgrade								
Veg Growing In Ballast/Roadway	9+96 - 14+15	B	Maint A	Spray Vegetation (Length)		419	TF	
Dirty (Fouled) Ballast	0+06 - 30+55	B	Maint A	Clean Ballast (Undercut)		3050	TF	
Insuff Crib Ballast (3" Ave Depth)	34+91 - 35+27	B	Maint A	Add Crib Ballast (3" Depth)		36	TF	
Dirty (Fouled) Ballast	30+55 - 35+27	B	Maint A	Clean Ballast (Undercut)		472	TF	
Drainage								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Geometry								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Appliances								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Turnout: 46								
Frog Flangeway Depth (Right)	18+67 - 19+65	N/A	Full Compl	Rebuild Frog	1.75	1	EA	
Frog Flangeway Depth (Left)	18+67 - 19+65	N/A	Full Compl	Rebuild Frog	1.75	1	EA	
Gauge At Frog Point (Right)	18+67 - 19+65	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.375	1		
Gauge At Switch Points	18+67 - 19+65	N/A	Full Compl	Correct Gauge	56.375	1	EA	
Rail Braces - Rigid (Left), Loose	18+67 - 19+65	N/A	Maint A	Tighten (Respoke) Rail Brace (Fixed)		1	EA	
Rail Braces - Rigid (Right), Loose	18+67 - 19+65	N/A	Maint A	Tighten (Respoke) Rail Brace (Fixed)		3	EA	
Surface and Alignment (Good)	18+67 - 19+65	N/A	Maint A	Do Nothing		1		
Continued on next page.								

Lead 3									
Area: Main Post				Segment Begin Location: 0+0.00				Track Category: A	
Track: Lead 3				Segment End Location: 35+27.27					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)	
Grade Crossings									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A	
Total Cost									

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Lower 4									
Area: Main Post				Segment Begin Location: 0+0.00				Track Category: A	
Track: Lower 4				Segment End Location: 16+39.86					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)	
Ties									
Single Defective Tie	0+00 - 16+40	N/A	Maint A	Install/Replace One Tie (7"x9"x8.5')		9	EA		
Rail									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	/A	
OTM									
Bolts, Missing/Cracked/Or Broken (1 Bolt)	0+00	B	Maint B	Install or Replace One Bolt		3	EA		
Ballast/Subgrade									
Dirty (Fouled) Ballast	0+02 - 16+40	B	Maint A	Clean Ballast (Undercut)		1638	TF		
Drainage									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	/A	
Geometry									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	/A	
Appliances									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	/A	
Turnouts									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	/A	
Grade Crossings									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	/A	
Total Cost									

Madigan Spur								
Area: Logistics			Segment Begin Location: 0+0.00			Track Category: B		
Track: Madigan Spur			Segment End Location: 47+20.67					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)
Ties								
Single Defective Tie	0+00 - 47+21	N/A	Maint A	Install/Replace One Tie (7"x9"x8.5')		56	EA	
Isolated Defective Tie Cluster (5 Ties)	0+00 - 47+21	N/A	Close to Traffic	Install/Replace Five Ties (7"x9"x8.5')		1	EA	
Rail								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
OTM								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Ballast/Subgrade								
Dirty (Fouled) Ballast	9+01 - 47+21	B	Maint A	Clean Ballast (Undercut)		3820	TF	
Dirty (Fouled) Ballast	0+05 - 9+01	B	Maint A	Clean Ballast (Undercut)		896	TF	
Drainage								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Geometry								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Appliances								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Turnouts								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Grade Crossings								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Total Cost								

Main								
Area: Logistics			Segment Begin Location: 0+0.00				Track Category: A	
Track: Main			Segment End Location: 52+46.69					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)
Ties								
Single Defective Tie	0+00 - 52+47	N/A	Maint A	Install/Replace One Tie (7"x9"x8.5')		27	EA	
Rail								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
OTM								
Bolts, Loose (1 Bolt) & Broken/Missing Washer	0+00	B	Maint B	Install or Replace One Bolt		2	EA	
Ballast/Subgrade								
Dirty (Fouled) Ballast	0+06 - 51+75	B	Maint A	Clean Ballast (Undercut)		5169	TF	
Drainage								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Geometry								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Appliances								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Turnout: 1								
Surface and Alignment (Good)	52+47 - 53+63	N/A	Maint A	Do Nothing		1		
Rail Braces - Rigid (Right), Loose	52+47 - 53+63	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		6	EA	
Gauge At Joints In Curved Closure Rail (1st)	52+47 - 53+63	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.375	1		
Gauge At Switch Points	52+47 - 53+63	N/A	Full Compl	Correct Gauge	56.375	1	EA	
Gauge At Joints In Curved Closure Rail (2nd)	52+47 - 53+63	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.625	1		
Turnout: 12								
Surface and Alignment (Good)	15+48 - 16+64	N/A	Maint A	Do Nothing		1		
Rail Braces - Rigid (Left), Loose	15+48 - 16+64	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		4	EA	
Rail Braces - Rigid (Right), Loose	15+48 - 16+64	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		4	EA	
Continued on next page.								

Main								
Area: Logistics			Segment Begin Location: 0+0.00				Track Category: A	
Track: Main			Segment End Location: 52+46.69					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)
Gauge At Switch Points	15+48 - 16+64	N/A	Full Compl	Correct Gauge	56.375	1	EA	
Turnout: 13								
Surface and Alignment (Good)	20+69 - 21+85	N/A	Maint A	Do Nothing		1		
Rail Braces - Rigid (Right), Loose	20+69 - 21+85	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		3	EA	
Gauge At Frog Point (Left)	20+69 - 21+85	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.25	1		
Gauge At Switch Points	20+69 - 21+85	N/A	Full Compl	Correct Gauge	56.375	1	EA	
Gauge At Frog Point (Right)	20+69 - 21+85	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.25	1		
Turnout: 19								
Rail Braces - Rigid (Left), Loose	34+63 - 35+79	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		3	EA	
Heel Joint Bolts (Right), Improper Size	34+63 - 35+79	N/A	Maint A	Install or Replace Heel Bolt Assembly		1	EA	
Rail Braces - Rigid (Right), Loose	34+63 - 35+79	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		3	EA	
Surface and Alignment (Good)	34+63 - 35+79	N/A	Maint A	Do Nothing		1		
Gauge At Switch Points	34+63 - 35+79	N/A	Full Compl	Correct Gauge	56.625	1	EA	
Guard Rail Flangeway Width (Right)	34+63 - 35+79	N/A	Full Compl	Tighten/Adjust Guard Rail Filler (Separator Block)	1.75	1	EA	
Gauge At Joints In Curved Closure Rail (2nd)	34+63 - 35+79	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.625	1		
Turnout: 2								
Rail Braces - Rigid (Right), Loose	1+16 - 0+00	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		6	EA	
Rail Braces - Rigid (Left), Loose	1+16 - 0+00	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		7	EA	
Surface and Alignment (Good)	1+16 - 0+00	N/A	Maint A	Do Nothing		1		
Continued on next page.								

Main								
Area: Logistics			Segment Begin Location: 0+0.00				Track Category: A	
Track: Main			Segment End Location: 52+46.69					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)
Gauge At Frog Point (Left)	1+16 - 0+00	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.75	1		
Gauge At Joints In Curved Closure Rail (2nd)	1+16 - 0+00	N/A	Full Compl	Correct Gauge	56.25	1	EA	
Turnout: 21								
Switch Rods, Loose	36+01 - 37+17	N/A	Close to Traffic	Tighten/Adjust Switch Rod		1	EA	
Surface and Alignment (Good)	36+01 - 37+17	N/A	Maint A	Do Nothing		1		
Rail Braces - Rigid (Left), Loose	36+01 - 37+17	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		5	EA	
Switch Ties, Defective (9 ft)	36+01 - 37+17	N/A	Maint A	Install/Replace 9' Switch Tie		1	EA	
Switch Ties, Defective (16 ft)	36+01 - 37+17	N/A	Maint A	Install/Replace 16' Switch Tie		2	EA	
Switch Ties, Defective (15 ft)	36+01 - 37+17	N/A	Maint A	Install/Replace 15' Switch Tie		1	EA	
Rail Braces - Rigid (Right), Loose	36+01 - 37+17	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		6	EA	
Gauge At Switch Points	36+01 - 37+17	N/A	Full Compl	Correct Gauge	56.75	1	EA	
Gauge At Frog Point (Right)	36+01 - 37+17	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.375	1		
Gauge At Frog Point (Left)	36+01 - 37+17	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.375	1		
Turnout: 26								
Surface and Alignment (Good)	40+69 - 41+85	N/A	Maint A	Do Nothing		1		
Rail Braces - Rigid (Left), Loose	40+69 - 41+85	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		2	EA	
Gauge At Joints In Curved Closure Rail (2nd)	40+69 - 41+85	N/A	Full Compl	Correct Gauge	56.75	1	EA	
Turnout: 3								
Surface and Alignment (Good)	3+20 - 4+36	N/A	Maint A	Do Nothing		1		
Switch Rods, Loose	3+20 - 4+36	N/A	Close to Traffic	Tighten/Adjust Switch Rod		1	EA	
Rail Braces - Rigid (Left), Loose	3+20 - 4+36	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		4	EA	
Rail Braces - Rigid (Right), Loose	3+20 - 4+36	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		7	EA	

Main									
Area: Logistics			Segment Begin Location: 0+0.00				Track Category: A		
Track: Main			Segment End Location: 52+46.69						
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM		Note(s)
Continued on next page.									
Gauge At Switch Points	3+20 - 4+36	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.25	1			
Gauge At Frog Point (Left)	3+20 - 4+36	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.25	1			
Gauge At Frog Point (Right)	3+20 - 4+36	N/A	10 MPH	Correct Gauge	56	1	EA		
Turnout: 33									
Rail Braces - Rigid (Right), Loose	44+53 - 45+69	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		1	EA		
Gauge At Frog Point (Left)	44+53 - 45+69	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.375	1			
Gauge At Joints In Curved Closure Rail (2nd)	44+53 - 45+69	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.75	1			
Gauge At Switch Points	44+53 - 45+69	N/A	Full Compl	Correct Gauge	56.75	1	EA		
Surface and Alignment (Good)	44+53 - 45+69	N/A	Maint A	Do Nothing		1			
Rail Braces - Rigid (Left), Loose	44+53 - 45+69	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		2	EA		
Turnout: 34									
Rail Braces - Rigid (Right), Loose	46+12 - 47+28	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		4	EA		
Rail Braces - Rigid (Left), Loose	46+12 - 47+28	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		4	EA		
Surface and Alignment (Good)	46+12 - 47+28	N/A	Maint A	Do Nothing		1			
Switch Ties, Defective (9 ft)	46+12 - 47+28	N/A	Maint A	Install/Replace 9' Switch Tie		2	EA		
Gauge At Joints In Curved Closure Rail (2nd)	46+12 - 47+28	N/A	Full Compl	Correct Gauge	56.625	1	EA		
Turnout: 5									
Surface and Alignment (Good)	6+06 - 7+22	N/A	Maint A	Do Nothing		1			
Gauge At Frog Point (Left)	6+06 - 7+22	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.875	1			
Continued on next page.									

Main								
Area: Logistics			Segment Begin Location: 0+0.00				Track Category: A	
Track: Main			Segment End Location: 52+46.69					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)
Rail Braces - Rigid (Left), Loose	6+06 - 7+22	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		1	EA	
Rail Braces - Rigid (Right), Loose	6+06 - 7+22	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		2	EA	
Gauge At Frog Point (Right)	6+06 - 7+22	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.875	1		
Gauge At Joints In Curved Closure Rail (2nd)	6+06 - 7+22	N/A	Full Compl	Correct Gauge	56.75	1	EA	
Grade Crossings								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A
Total Cost								

North 1									
Area: Logistics				Segment Begin Location: 0+0.00				Track Category: B	
Track: North 1				Segment End Location: 23+84.37					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)	
Ties									
Single Defective Tie	0+00 - 23+84	N/A	Maint A	Install/Replace One Tie (7"x9"x8.5')		1	EA		
Rail									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A	
OTM									
Bolts, Loose (1 Bolt) & Broken/Missing Washer	0+00	B	Maint B	Install or Replace One Bolt		1	EA		
Ballast/Subgrade									
Dirty (Fouled) Ballast	4+73 - 22+89	B	Maint A	Clean Ballast (Undercut)		1817	TF		
Drainage									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A	
Geometry									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A	
Appliances									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A	
Turnouts									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A	
Grade Crossings									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A	
Total Cost									

North 2								
Area: Logistics			Segment Begin Location: 0+0.00			Track Category: B		
Track: North 2			Segment End Location: 26+16.09					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)
Ties								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A
Rail								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A
OTM								
Bolts, Loose (1 Bolt) & Broken/Missing Washer	0+00	B	Maint B	Install or Replace One Bolt		1	EA	
Ballast/Subgrade								
Dirty (Fouled) Ballast	0+08 - 26+16	B	Maint A	Clean Ballast (Undercut)		2608	TF	
Drainage								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A
Geometry								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A
Appliances								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A
Turnout: 16								
Frog Flangeway Depth (Left)	23+84 - 25+00	N/A	Full Compl	Rebuild Frog	1.75	1	EA	
Frog Flangeway Depth (Right)	23+84 - 25+00	N/A	Full Compl	Rebuild Frog	1.75	1	EA	
Surface and Alignment (Good)	23+84 - 25+00	N/A	Maint A	Do Nothing		1		
Switch Ties, Defective (13 ft)	23+84 - 25+00	N/A	Maint A	Install/Replace 13' Switch Tie		1	EA	
Rail Braces - Rigid (Right), Loose	23+84 - 25+00	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		2	EA	
Rail Braces - Rigid (Left), Loose	23+84 - 25+00	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		3	EA	
Turnout: 9								
Frog Flangeway Depth (Left)	1+15 - 2+31	N/A	Full Compl	Rebuild Frog	1.75	1	EA	
Frog Flangeway Depth (Right)	1+15 - 2+31	N/A	Full Compl	Rebuild Frog	1.75	1	EA	
Surface and Alignment (Good)	1+15 - 2+31	N/A	Maint A	Do Nothing		1		
Continued on next page.								

North 2									
Area: Logistics				Segment Begin Location: 0+0.00				Track Category: B	
Track: North 2				Segment End Location: 26+16.09					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)	
Grade Crossings									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A	
Total Cost									

North 3								
Area: Logistics			Segment Begin Location: 0+0.00			Track Category: B		
Track: North 3			Segment End Location: 29+82.49					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)
Ties								
Single Defective Tie	0+00 - 29+82	N/A	Maint A	Install/Replace One Tie (7"x9"x8.5')		1	EA	
Rail								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A
OTM								
Bolts, Loose (1 Bolt) & Broken/Missing Washer	0+00	B	Maint B	Install or Replace One Bolt		5	EA	
Ballast/Subgrade								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A
Drainage								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A
Geometry								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A
Appliances								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A
Turnout: 17								
Gauge At Frog Point (Left)	26+85 - 28+01	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.25	1		
Gauge At Joints In Curved Closure Rail (2nd)	26+85 - 28+01	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.625	1		
Gauge At Switch Points	26+85 - 28+01	N/A	Full Compl	Correct Gauge	56.875	1	EA	
Rail Braces - Rigid (Right), Loose	26+85 - 28+01	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		4	EA	
Surface and Alignment (Good)	26+85 - 28+01	N/A	Maint A	Do Nothing		1		
Turnout: 8								
Switch Ties, Defective (9 ft)	1+82 - 2+98	N/A	Maint A	Install/Replace 9' Switch Tie		1	EA	
Rail Braces - Rigid (Right), Loose	1+82 - 2+98	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		1	EA	
Rail Braces - Rigid (Left), Loose	1+82 - 2+98	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		4	EA	
Continued on next page.								

North 3									
Area: Logistics				Segment Begin Location: 0+0.00				Track Category: B	
Track: North 3				Segment End Location: 29+82.49					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM		Note(s)
Surface and Alignment (Good)	1+82 - 2+98	N/A	Maint A	Do Nothing		1			
Grade Crossings									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		A
Total Cost									

North Wye								
Area: ASP			Segment Begin Location: 0+0.00			Track Category: B		
Track: North Wye			Segment End Location: 18+62.94					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)
Ties								
Single Defective Tie	0+00 - 18+63	N/A	Maint A	Install/Replace One Tie (7"x9"x8.5')		16	EA	
Rail								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	/A
OTM								
Bolts, Loose (1 Bolt) & Broken/Missing Washer	0+00	B	Maint B	Install or Replace One Bolt		1	EA	
Ballast/Subgrade								
Dirty (Fouled) Ballast	0+36 - 16+93	B	Maint A	Clean Ballast (Undercut)		1657	TF	
Drainage								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	/A
Geometry								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	/A
Appliances								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	/A
Turnouts								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	/A
Grade Crossings								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	/A
Total Cost								

South 1								
Area: Logistics			Segment Begin Location: 0+0.00			Track Category: A		
Track: South 1			Segment End Location: 26+12.73					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)
Ties								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A
Rail								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A
OTM								
Bolts, Loose (1 Bolt) & Broken/Missing Washer	0+00	B	Maint B	Install or Replace One Bolt		3	EA	
Ballast/Subgrade								
Dirty (Fouled) Ballast	18+09 - 26+13	B	Maint A	Clean Ballast (Undercut)		804	TF	
Drainage								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A
Geometry								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A
Appliances								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A
Turnouts								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A
Grade Crossings								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A
Total Cost								

South 2								
Area: Logistics			Segment Begin Location: 0+0.00			Track Category: B		
Track: South 2			Segment End Location: 23+13.49					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)
Ties								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	/A
Rail								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	/A
OTM								
Bolts, Loose (1 Bolt) & Broken/Missing Washer	0+00	B	Maint B	Install or Replace One Bolt		1	EA	
Ballast/Subgrade								
Dirty (Fouled) Ballast	21+89 - 23+13	B	Maint A	Clean Ballast (Undercut)		124	TF	
Dirty (Fouled) Ballast	1+23 - 8+83	B	Maint A	Clean Ballast (Undercut)		760	TF	
Drainage								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	/A
Geometry								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	/A
Appliances								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	/A
Turnouts								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	/A
Grade Crossings								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	/A
Total Cost								

South 3								
Area: Logistics			Segment Begin Location: 0+0.00			Track Category: B		
Track: South 3			Segment End Location: 29+82.90					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)
Ties								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Rail								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
OTM								
Bolts, Loose (1 Bolt) & Broken/Missing Washer	0+00	B	Maint B	Install or Replace One Bolt		1	EA	
Ballast/Subgrade								
Dirty (Fouled) Ballast	3+20 - 26+96	B	Maint A	Clean Ballast (Undercut)		2375	TF	
Drainage								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Geometry								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Appliances								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Turnout: 10								
Rail Braces - Rigid (Right), Loose	3+31 - 4+47	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		2	EA	
Rail Braces - Rigid (Left), Loose	3+31 - 4+47	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		2	EA	
Switch Ties, Defective (15 ft)	3+31 - 4+47	N/A	Maint A	Install/Replace 15' Switch Tie		1	EA	
Surface and Alignment (Good)	3+31 - 4+47	N/A	Maint A	Do Nothing		1		
Guard Rail Flangeway Width (Right)	3+31 - 4+47	N/A	Full Compl	Tighten/Adjust Guard Rail Filler (Separator Block)	1.75	1	EA	
Gauge At Frog Point (Right)	3+31 - 4+47	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.375	1		
Gauge At Frog Point (Left)	3+31 - 4+47	N/A	Full Compl	Correct Gauge	56.625	1	EA	
Turnout: 15								
Rail Braces - Rigid (Right), Loose	25+33 - 26+49	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		2	EA	
Continued on next page.								

South 3								
Area: Logistics Track: South 3			Segment Begin Location: 0+0.00 Segment End Location: 29+82.90			Track Category: B		
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)
Gauge At Frog Point (Left)	25+33 - 26+49	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.75	1		
Surface and Alignment (Good)	25+33 - 26+49	N/A	Maint A	Do Nothing		1		
Gauge At Joints In Curved Closure Rail (1st)	25+33 - 26+49	N/A	Full Compl	Correct Gauge	56.375	1	EA	
Turnout: 18								
Frog Bolts, Loose	26+86 - 28+02	N/A	Maint A	Tighten Frog Bolt		1	EA	
Surface and Alignment (Good)	26+86 - 28+02	N/A	Maint A	Do Nothing		1		
Gauge At Joints In Curved Closure Rail (2nd)	26+86 - 28+02	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.75	1		
Gauge At Switch Points	26+86 - 28+02	N/A	Full Compl	Correct Gauge	56.375	1	EA	
Rail Braces - Rigid (Left), Loose	26+86 - 28+02	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		1	EA	
Turnout: 7								
Heel Joint Bolts (Left), Loose	1+82 - 2+98	N/A	Maint A	Tighten Heel Bolts		1	EA	
Heel Joint Bolts (Right), Loose	1+82 - 2+98	N/A	Maint A	Tighten Heel Bolts		1	EA	
Rail Braces - Rigid (Right), Loose	1+82 - 2+98	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		3	EA	
Surface and Alignment (Good)	1+82 - 2+98	N/A	Maint A	Do Nothing		1		
Gauge At Frog Point (Right)	1+82 - 2+98	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.375	1		
Gauge At Joints In Curved Closure Rail (2nd)	1+82 - 2+98	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.25	1		
Gauge At Switch Points	1+82 - 2+98	N/A	Full Compl	Correct Gauge	56.25	1	EA	
Grade Crossings								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A
Total Cost								

South Wye								
Area: ASP			Segment Begin Location: 0+0.00			Track Category: B		
Track: South Wye			Segment End Location: 18+39.61					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)
Ties								
Single Defective Tie	0+00 - 18+40	N/A	Maint A	Install/Replace One Tie (7"x9"x8.5')		19	EA	
Rail								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
OTM								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Ballast/Subgrade								
Dirty (Fouled) Ballast	1+31 - 16+72	B	Maint A	Clean Ballast (Undercut)		1541	TF	
Drainage								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Geometry								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Appliances								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Turnouts								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Grade Crossings								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	
Total Cost								

Track 1								
Area: Logistics			Segment Begin Location: 0+0.00			Track Category: B		
Track: Track 1			Segment End Location: 30+5.75					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)
Ties								
Single Defective Tie	0+00 - 30+06	N/A	Maint A	Install/Replace One Tie (7"x9"x8.5')		3	EA	
Rail								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A
OTM								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A
Ballast/Subgrade								
Dirty (Fouled) Ballast	0+03 - 30+06	B	Maint A	Clean Ballast (Undercut)		3003	TF	
Pumping Ties (Both Ends)	28+49 - 30+06	B	Maint A	Excavate Mud Crib		157	TF	
Drainage								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A
Geometry								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A
Appliances								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A
Turnouts								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A
Grade Crossings								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A
Total Cost								

Track 2								
Area: Logistics			Segment Begin Location: 0+0.00			Track Category: B		
Track: Track 2			Segment End Location: 30+98.63					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)
Ties								
Single Defective Tie	0+00 - 30+99	N/A	Maint A	Install/Replace One Tie (7"x9"x8.5')		2	EA	
Rail								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	/A
OTM								
Bolts, Loose (1 Bolt) & Broken/Missing Washer	0+00	B	Maint B	Install or Replace One Bolt		1	EA	
Ballast/Subgrade								
Dirty (Fouled) Ballast	0+01 - 30+99	B	Maint A	Clean Ballast (Undercut)		3098	TF	
Pumping Ties (Both Ends)	1+05 - 1+38	B	Maint A	Excavate Mud Crib		33	TF	
Drainage								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	/A
Geometry								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	/A
Appliances								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	/A
Turnout: 32								
Frog Flangeway Width (Right)	7+18 - 8+34	N/A	Full Compl	Grind and Dress Frog	1.75	1	EA	
Frog Flangeway Depth (Right)	7+18 - 8+34	N/A	Full Compl	Rebuild Frog	1.75	1	EA	
Frog Flangeway Width (Left)	7+18 - 8+34	N/A	Full Compl	Grind and Dress Frog	1.75	1	EA	
Frog Flangeway Depth (Left)	7+18 - 8+34	N/A	Full Compl	Rebuild Frog	1.75	1	EA	
Rail Braces - Rigid (Right), Loose	7+18 - 8+34	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		2	EA	
Rail Braces - Rigid (Left), Loose	7+18 - 8+34	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		2	EA	
Switch Ties, Defective (9 ft)	7+18 - 8+34	N/A	Maint A	Install/Replace 9' Switch Tie		1	EA	
Switch Ties, Defective (12 ft)	7+18 - 8+34	N/A	Maint A	Install/Replace 12' Switch Tie		1	EA	
Surface and Alignment (Good)	7+18 - 8+34	N/A	Maint A	Do Nothing		1		
Continued on next page.								

Track 2									
Area: Logistics				Segment Begin Location: 0+0.00				Track Category: B	
Track: Track 2				Segment End Location: 30+98.63					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM		Note(s)
Switch Point Gap (Left)	7+18 - 8+34	N/A	Maint B	Adjust and Lubricate Switch	0.125	1	EA		
Grade Crossings									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		N/A
Total Cost									

Track 3									
Area: Logistics				Segment Begin Location: 0+0.00				Track Category: B	
Track: Track 3				Segment End Location: 23+79.47					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)	
Ties									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A	
Rail									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A	
OTM									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A	
Ballast/Subgrade									
Dirty (Fouled) Ballast	23+79 - 23+78	B	Maint A	Clean Ballast (Undercut)		1	TF		
Drainage									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A	
Geometry									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A	
Appliances									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A	
Turnouts									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A	
Grade Crossings									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A	
Total Cost									

Track 4								
Area: Logistics			Segment Begin Location: 0+0.00			Track Category: B		
Track: Track 4			Segment End Location: 31+63.28					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)
Ties								
Single Defective Tie	0+00 - 31+63	N/A	Maint A	Install/Replace One Tie (7"x9"x8.5')		7	EA	
Rail								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A
OTM								
Bolts, Loose (1 Bolt) & Broken/Missing Washer	0+00	B	Maint B	Install or Replace One Bolt		1	EA	
Ballast/Subgrade								
Dirty (Fouled) Ballast	31+61 - 31+63	B	Maint A	Clean Ballast (Undercut)		2	TF	
Pumping Ties (Both Ends)	4+47 - 31+63	B	Maint A	Excavate Mud Crib		2,716	TF	
Drainage								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A
Geometry								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A
Appliances								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A
Turnout: 31								
Rail Braces - Rigid (Right), Loose	7+51 - 8+67	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		6	EA	
Surface and Alignment (Good)	7+51 - 8+67	N/A	Maint A	Do Nothing		1		
Rail Braces - Rigid (Left), Loose	7+51 - 8+67	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		2	EA	
Grade Crossings								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A
Total Cost								

Track 5								
Area: Logistics			Segment Begin Location: 0+0.00			Track Category: A		
Track: Track 5			Segment End Location: 24+11.24					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)
Ties								
Single Defective Tie	0+00 - 24+11	N/A	Maint A	Install/Replace One Tie (7"x9"x8.5')		1	EA	
Rail								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A
OTM								
Rail End Mis (Tread) > 0.125" & <= 0.1875" (>1/8 - <=3/16)	23+61	R	Maint B	Eliminate Rail Mismatch		1	EA	
Bolts, Loose (1 Bolt) & Broken/Missing Washer	0+00	B	Maint B	Install or Replace One Bolt		3	EA	
Ballast/Subgrade								
Dirty (Fouled) Ballast	24+10 - 24+11	B	Maint A	Clean Ballast (Undercut)		2	TF	
Drainage								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A
Geometry								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A
Appliances								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A
Turnouts								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A
Grade Crossings								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A
Total Cost								

Track 6									
Area: Logistics				Segment Begin Location: 0+0.00				Track Category: A	
Track: Track 6				Segment End Location: 32+71.02					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM		Note(s)
Ties									
Single Defective Tie	0+00 - 32+71	N/A	Maint A	Install/Replace One Tie (7"x9"x8.5')		4	EA		
Rail									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		I/A
OTM									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		I/A
Ballast/Subgrade									
Dirty (Fouled) Ballast	4+45 - 32+71	B	Maint A	Clean Ballast (Undercut)		2826	TF		
Drainage									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		I/A
Geometry									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		I/A
Appliances									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		I/A
Turnout: 24									
Surface and Alignment (Good)	1+16 - 2+32	N/A	Maint A	Do Nothing		1			
Rail Braces - Rigid (Right), Loose	1+16 - 2+32	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		6	EA		
Rail Braces - Rigid (Left), Loose	1+16 - 2+32	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		6	EA		
Gauge At Switch Points	1+16 - 2+32	N/A	Full Compl	Correct Gauge	56.375	1	EA		
Gauge At Joints In Curved Closure Rail (1st)	1+16 - 2+32	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.125	1			
Gauge At Frog Point (Left)	1+16 - 2+32	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.75	1			
Gauge At Frog Point (Right)	1+16 - 2+32	N/A	Full Compl	Corrected by Other Work – Correct Gauge	57	1			
Gauge At Joints In Curved Closure Rail (2nd)	1+16 - 2+32	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.75	1			
Turnout: 30									
Continued on next page.									

Track 6									
Area: Logistics Track: Track 6			Segment Begin Location: 0+0.00 Segment End Location: 32+71.02				Track Category: A		
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM		Note(s)
Surface and Alignment (Good)	8+27 - 9+43	N/A	Maint A	Do Nothing		1			
Rail Braces - Rigid (Left), Loose	8+27 - 9+43	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		4	EA		
Switch Rods, Loose	8+27 - 9+43	N/A	Close to Traffic	Tighten/Adjust Switch Rod		1	EA		
Cotter Keys, Missing	8+27 - 9+43	N/A	10 MPH	Install Cotter Key		1	EA		
Gauge At Joints In Curved Closure Rail (2nd)	8+27 - 9+43	N/A	Full Compl	Correct Gauge	56.75	1	EA		
Gauge At Frog Point (Left)	8+27 - 9+43	N/A	Full Compl	Corrected by Other Work – Correct Gauge	56.875	1			
Gauge At Frog Point (Right)	8+27 - 9+43	N/A	Full Compl	Corrected by Other Work – Correct Gauge	57	1			
Rail Braces - Rigid (Right), Loose	8+27 - 9+43	N/A	Maint A	Tighten (Respike) Rail Brace (Fixed)		6	EA		
Grade Crossings									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		/A
Total Cost									

Track 7								
Area: Logistics			Segment Begin Location: 0+0.00			Track Category: A		
Track: Track 7			Segment End Location: 24+44.58					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)
Ties								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A
Rail								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A
OTM								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A
Ballast/Subgrade								
Low or Dropped Tie	24+26 - 24+28	R	Maint A	Tamp Ballast Around Tie & Tighten Hold Down		1	EA	
Dirty (Fouled) Ballast	0+02 - 24+42	B	Maint A	Clean Ballast (Undercut)		2441	TF	
Drainage								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A
Geometry								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A
Appliances								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A
Turnouts								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A
Grade Crossings								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A
Total Cost								

Track 8								
Area: Logistics			Segment Begin Location: 0+0.00			Track Category: A		
Track: Track 8			Segment End Location: 30+95.50					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)
Ties								
Single Defective Tie	0+00 - 30+96	N/A	Maint A	Install/Replace One Tie (7"x9"x8.5')		1	EA	
Rail								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A
OTM								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A
Ballast/Subgrade								
Dirty (Fouled) Ballast	26+55 - 30+92	B	Maint A	Clean Ballast (Undercut)		438	TF	
Dirty (Fouled) Ballast	4+48 - 26+55	B	Maint A	Clean Ballast (Undercut)		2207	TF	
Drainage								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A
Geometry								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A
Appliances								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A
Turnouts								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A
Grade Crossings								
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	N/A
Total Cost								

Upper 4									
Area: Logistics				Segment Begin Location: 0+0.00				Track Category: A	
Track: Upper 4				Segment End Location: 11+13.74					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM	Note(s)	
Ties									
Single Defective Tie	0+00 - 11+14	N/A	Maint A	Install/Replace One Tie (7"x9"x8.5')		6	EA		
Rail									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A	
OTM									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A	
Ballast/Subgrade									
Dirty (Fouled) Ballast	2+49 - 11+14	B	Maint A	Clean Ballast (Undercut)		865	TF		
Drainage									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A	
Geometry									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A	
Appliances									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A	
Turnouts									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A	
Grade Crossings									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A	A	
Total Cost									

Upper 6									
Area: Logistics				Segment Begin Location: 0+0.00				Track Category: A	
Track: Upper 6				Segment End Location: 19+55.57					
Defect	Station (ft)	Rail	Standard	Work Action	Meas (in.)	Quant	UM		Note(s)
Ties									
Single Defective Tie	0+00 - 19+56	N/A	Maint A	Install/Replace One Tie (7"x9"x8.5')		11	EA		
Rail									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		N/A
OTM									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		N/A
Ballast/Subgrade									
Dirty (Fouled) Ballast	13+47 - 19+55	B	Maint A	Clean Ballast (Undercut)		608	TF		
Drainage									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		N/A
Geometry									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		N/A
Appliances									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		N/A
Turnouts									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		N/A
Grade Crossings									
Defect Free	N/A	N/A	N/A	N/A		N/A	N/A		N/A
Total Cost									

